Spatially Anisotropic Reasoning Network for Remote Sensing Scene Graph Generation

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***Abstract*—Remote sensing scene graph generation (RS-SGG) aims to advance remote sensing image interpretation from primitive entity recognition to high-level holistic scene understanding. Due to the large spatial coverage of remote sensing images, objects are often organized into multiple functional subscenes, while predicate semantics are highly dependent on the spatial configurations of the subject and object. However, existing scene graph generation (SGG) methods typically perform unconstrained global interactions over the entire panoramic scene, which introduces substantial noise in multisubscene scenarios, and they rarely exploit the relative spatial structure underlying subject–object relations. To address these issues, we propose the spatially anisotropic reasoning network (SARNet) for RS-SGG. First, we design an anisotropic elliptical influence propagation (AEIP) module that models an adaptive spatial influence region for each object using anisotropic Gaussian ellipses derived from oriented bounding boxes (OBBs), focusing on instance-centered interactions while suppressing noisy long-range relations. Second, we introduce a relative spatial configuration-guided attention (RSCA) module that incorporates the relative geometric configuration between each relation and its corresponding subject–object pair, and reallocates channel-wise attention to enhance spatially discriminative features for predicate prediction. Extensive experiments on the STAR and AUG benchmarks demonstrate that the proposed method achieves state-of-the-art performances on RS-SGG tasks, highlighting the importance of spatially structured reasoning that jointly captures object-centered functional context and relative spatial configurations in large-area remote sensing images. The code is publicly available at https://github.com/Bamboo0216/SARNet**

***Index Terms*—Anisotropic Gaussian ellipse, remote sensing, scene graph generation (SGG), spatial configuration.**

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I. Introduction

REMOTE sensing image interpretation has achieved remarkable progress, in the field of object detection [2], [3], [4], change detection [5], [6], [7], [8], semantic segmentation [9], [10], [11], and instance segmentation [12], [13]. However, remote sensing scenes exhibit strong structural dependency, where objects and regions are organized in correlated spatial patterns rather than existing independently. Scene graph [14], [15] has been used for explaining and reasoning over complex visual scenes, where the detected objects and their pairwise relationships in an image are denoted as nodes and edges, respectively, transforming unstructured semantics into a structured graph. The scene graph provides an important intermediate representation for high-level cognition, and has also been increasingly applied to multimodal large language model (MLLM)-based spatial perception and reasoning [16], [17], [18]. Remote sensing scene graph generation (RS-SGG) is expected to advance remote sensing image interpretation from isolated entity perception to holistic scene-level understanding.

Despite the potential of the scene graph for comprehensive remote sensing scene understanding, the unique properties of remote sensing imagery introduce additional challenges that must be addressed for accurate relational modeling in RS-SGG. High-resolution remote sensing images usually cover extremely large geographic areas and contain a massive number of objects distributed over wide scenes. Due to the large spatial extent, objects in remote sensing imagery are rarely uniformly scattered. Instead, objects with similar functions tend to cluster together, forming spatially coherent regions. These regions can be naturally regarded as **subscenes** as highlighted by the yellow dashed boxes in Fig. 1(a), which represent loosely bounded functional areas formed by spatially proximate objects with related semantics. Generally, relations are mostly localized within each subscene. For example, in a port city remote sensing scene [see Fig. 1(a)], relations within

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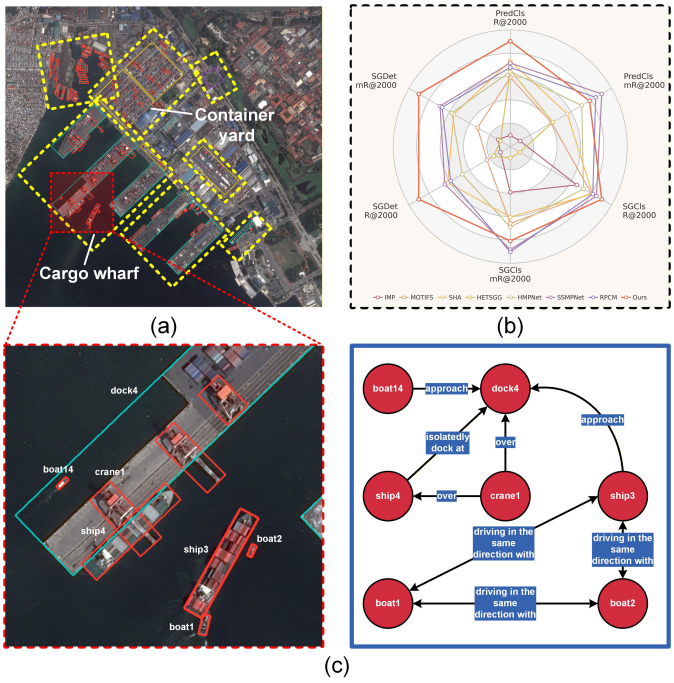


Fig. 1. (a) Single large-area remote sensing image may contain multiple spatially coherent regions (denoted by the yellow dashed boxes) corresponding to different functional areas. (b) Performance comparison between our model and other methods on different evaluation tasks of the benchmarking STAR [1] dataset. (c) Details of the cargo wharf (left) and its corresponding scene graph (right). Predicate inference between two objects relies heavily on reasoning within the local subscene and its spatial configuration.

the cargo wharf mainly occur between the ships and the wharf, while relations within the container yard mostly involve the containers and vehicles. Cross-subscene relations are rarely meaningful; for instance, the relation between a ship and a distant container is usually negligible, as they belong to different spatial regions with weak semantic association. As pointed out in prior studies [19], relational reasoning should prioritize the most informative interactions. When excessive unconstrained global interactions are involved, weak associations across distant regions may obscure the object-centered local context that is essential for relation prediction. Therefore, effective relational reasoning in remote sensing scenes requires object representations with stronger awareness of their surrounding subscene-like context, while indiscriminate global interactions across distant regions may introduce unnecessary noise.

Another distinctive characteristic of remote sensing imagery is that objects exhibit large-scale variations but lack rich appearance cues, since height, depth, and fine-grained texture information are often unavailable. As a result, semantic relations rely more heavily on spatial configurations than on visual appearance. Many predicates, such as relative movement, alignment, overlap, or isolation, are fundamentally determined by geometric layout and relative positions rather than object categories alone, as illustrated in Fig. 1(c). Therefore, accurate relation inference in remote sensing scenes requires explicit modeling of spatial geometry and structural configurations between objects.

Existing SGG research has been mainly developed on natural images [20], [21], [22], [23], [24], [25], [26], giving rise to a variety of relation reasoning frameworks, including message passing networks [20] and Transformers [27]. In contrast to remote sensing imagery, natural images usually depict compact scenes with richer appearance cues, where objects are densely arranged and relations are less strictly constrained by spatial layout, making global contextual modeling a reasonable assumption. Such assumptions do not hold in remote sensing imagery as discussed above. Consequently, existing SGG models often lack selective attention to local environments and fail to explicitly characterize the spatial geometry within subject–object relations, leading to degraded performance in remote sensing scenarios.

To address these issues, we propose the spatially anisotropic reasoning network (SARNet). Inspired by the observation that meaningful relations in remote sensing images are often concentrated within functional regions, we design an anisotropic elliptical influence propagation (AEIP) to regulate relational interactions from an object-centered perspective. Since such subscene-like regions usually lack explicit annotations and strictly defined boundaries, directly segmenting subscenes or assigning objects to them can be ambiguous. Therefore, instead of explicitly modeling subscenes, AEIP represents the local functional neighborhood of each object using a soft anisotropic elliptical influence region. Specifically, each object is assigned a diffusion Gaussian ellipse whose shape and scale are adaptively determined by its geometry and semantics, approximating the spatial extent over which the object is likely to interact with its surroundings. The overlap between influence regions defines the propagation strength between objects and relations, allowing the model to emphasize nearby and structurally related instances while suppressing weak interactions across distant regions. In this way, AEIP encourages relation reasoning to focus on object-centered local structural clusters that are consistent with the subscene-like organization observed in remote sensing scenes, without requiring explicit subscene segmentation or hard boundary definitions. As a result, SARNet strengthens meaningful local relational context while reducing noisy long-range interactions in large-area remote sensing images.

Meanwhile, to address the strong dependence of relational semantics on spatial configuration in remote sensing imagery, we design a relative spatial configuration-guided attention (RSCA) module that explicitly models the relative layout between the subject, object, and relation. Specifically, RSCA incorporates relative geometric information between each relation and its corresponding subject and object, and uses them to guide the attention reallocation over the subject and object features to emphasize spatially discriminative cues for relation inference. By injecting this subject–object relative spatial configuration, the model can better capture geometry-dominated predicates that frequently occur in remote sensing scenes.

Overall, our contributions are threefold as follows.

1) We propose the AEIP module, which constrains relational reasoning through object-centered spatial influence modeling inspired by the subscene-like organization of remote sensing images. By assigning each object a soft anisotropic elliptical influence region, AEIP captures local functional associations without requiring

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explicit subscene segmentation or hard boundary definitions, thereby suppressing noisy long-range interactions in large-area remote sensing scenes.

2) We design the RSCA module, which generates channel-wise attention weights for subject and object features based on their relative spatial configuration with respect to each relation. RSCA emphasizes geometry-sensitive and relation-discriminative cues, thereby improving the modeling of structure-dependent predicates under complex spatial layouts in remote sensing imagery.
3) Extensive experiments on the STAR [1] and AUG [28] benchmarks demonstrate that the proposed SARNet achieves state-of-the-art performance, especially under the most challenging SGDet task as illustrated in Fig. 1(b). More importantly, the results show that explicitly modeling spatial influence and relative spatial configuration is essential for accurate RS-SGG, highlighting the necessity of designing relation reasoning frameworks tailored to the unique spatial characteristics of remote sensing scenes.

II. Related Work

A. Natural Image Scene Graph Generation (SGG)

Mainstream SGG methods typically treat objects and relations as basic reasoning units, and perform a higher-order message passing process among these units to capture rich contextual information for relation inference. Diverse SGG architectures have been proposed [20], [29], [30], [31], [32], [33], [34], [35], [36], with two dominant frameworks: message passing neural networks (MPNNs) and Transformers. Message passing enables flexible contextual modeling by propagating information among objects. GPS-Net [30] enables direction-specific message propagation between objects. BGNN [31] suppresses unreliable message passing using confidence-based filtering. HetSGG [32] performs relation-aware message passing on heterogeneous graphs to capture semantic heterogeneity. On the other hand, Transformers achieve high-quality relation modeling due to their self-attention (SA) mechanism, which allows for global context aggregation and flexible modeling of interobject dependencies. SHA [34] fuses visual and semantic cues via a dual-stream attention design, while SQUAT [36] performs contextual reasoning through edge-to-node and edge-to-edge attentions for relation features, as well as node-to-node and node-to-edge attentions for object features. Besides, some studies [37] focus on addressing the long-tail distributions in SGG. Although existing SGG methods have explored various context construction designs, they do not fully adapt to the unique characteristics of remote sensing scenes. Remote sensing images often exhibit substantial scale variation and complex spatial configurations across large-area scenes. Natural image SGG (NI-SGG) methods typically lack targeted modeling for these challenges, and thus show limited performance when handling massive and complex spatial information in remote sensing images.

B. Remote Sensing SGG

RS-SGG is still at an early stage. Benchmarks for RS-SGG are still less comprehensive than those in more established remote sensing tasks, such as object detection [38] and scene classification [39]. Chen et al. [40] proposed the GRTRD dataset, where geospatial relations are mainly represented by topology, direction, and distance. Li et al. [41] propose the RSSGD dataset built on descriptive sentences from the remote sensing caption dataset RSICD [42], and introduce an attention-based dilated-convolution graph convolution network for multiscale semantic fusion. SRSG [43] focuses on building high-quality scene graphs with object morphology, introduces the segmentation-based dataset S2SG, and uses pixel coordinates from segmentation results to encode object shapes and relative spatial positions for predicate prediction. Overall, early RS-SGG datasets are small in scale and limited in annotation richness, containing only a restricted number of object and predicate categories. This lack of diversity and coverage makes it difficult to train models that generalize well to complex and diverse remote sensing scenes.

The emergence of large-scale relation-annotated datasets, such as STAR [1], AUG [28], and ReCon1M [44], marks a new stage toward practical applications. STAR establishes a large-scale benchmark for SGG on high-resolution satellite images, aiming to jointly model multiscale objects and their semantic relations in complex remote sensing scenes. It covers global representative human activity scenes; each image contains multiple subscenes, such as airports, ports, nuclear power, and plant areas.

Driven by the development of the benchmarks, recent models have explored stronger architectures for RS-SGG. EGCAN [44] employs a strong Transformer module and encodes all object features as a sequence via SA. RPCM [1] performs progressive global context-aware message passing based on prototype guidance to avoid confusion from excessive global context. HMPNet [45] explicitly models higher-order semantic associations among predicate prototypes using a multimodal hypergraph, improving prototype matching. To mitigate the long-tail distribution problem of the dataset [46], a hierarchical prototype learning network [47] is proposed to enhance fine-grained predicate prediction via an aggregation-decomposition strategy.

Among existing RS-SGG methods, SSMPNet [48] is closely related to our motivation, as it also considers the coexistence of multiple scenes and the interference caused by unconstrained global interactions. However, SSMPNet addresses this issue from a *prior-driven scene-category decoupling* perspective: it maps objects and relation pairs to predefined scene types according to object labels, and then performs scene-aware interaction and prototype matching within this discrete semantic taxonomy. Such priors are useful but may be insufficient to capture image-specific spatial associations, since objects assigned to the same scene type can be spatially distant and weakly correlated, while nearby objects from different scene types may still form meaningful relations. In contrast, SARNet does not rely on predefined scene assignments or hard subscene boundaries, but models relational context according to the actual geometry of each image. Specifically, AEIP constructs object-centered anisotropic influence regions and uses their spatial overlap to adaptively regulate interaction strength, shifting the reasoning paradigm from category-defined scene

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grouping to *geometry-driven local interaction modeling*. In addition, RSCA further refines predicate prediction by explicitly exploiting relative subject–object spatial configurations, which is orthogonal to the scene-specific prototype learning strategy in SSMPNet.

In general, existing methods still insufficiently exploit the image-specific spatial organization of objects when modeling relational context, and rarely characterize the relative spatial configuration within each subject–object relation. This limitation is particularly critical in remote sensing images, where meaningful relations are often shaped by local functional associations and spatial layouts rather than by object categories alone. Our method addresses this issue by modeling object-centered spatial influence to regulate local relational propagation, while explicitly incorporating subject–object spatial configuration during predicate reasoning.

C. Gaussian Oriented Bounding Boxes (GBBs)

GBBs [49], [50], [51], [52], [53], [54], [55] have been widely adopted in oriented object detection, acting as an intermediate representation and mapping a rotated box into a 2-D Gaussian ellipse. A series of studies primarily focus on applying GBB to the loss computation to better approximate rotated IoU and improve optimization stability. KFIoU [51] leverages Gaussian product to emulate the computation of rotated IoU, achieving competitive performance without additional hyperparameters. Zhou et al. [54] propose a linear GBB representation, preserving continuity while markedly improving numerical stability during training. In this work, we present a novel application of GBB in RS-SGG. We posit that GBB facilitates explicit modeling of each object's influence on its surroundings. We model the influence field of each object as an adaptively expanded anisotropic Gaussian ellipse derived from its oriented bounding box (OBB), whose strength decays with distance and direction, providing principled interaction weights for spatially aware relation reasoning.

III. Methodology

This section presents the proposed SARNet that follows the widely used two-stage pipeline to address the task of RS-SGG. In the following sections, we first briefly introduce the overall framework of SARNet in Section III-A, then detail the AEIP module which emphasizes object-centered spatial influence context in Section III-B, and the RSCA module that models the subject–object spatial configuration for predicate reasoning in Section III-C.

A. Overall Framework

SARNet follows the widely adopted two-stage pipeline for RS-SGG. In the first stage, a pretrained oriented detector (e.g., HOD-Net [1]) is applied to detect objects, each of which contains the classification scores, an OBB, and the visual feature. The feature of a relation between two objects is pooled from the union OBB of them over the whole-image feature map. In the second stage, as shown in the upper middle part of Fig. 2, SARNet performs relation reasoning with three components: 1) an *object enhancer* that refines object representations and updates object predictions; 2) a *relation encoder* that first captures global interactions via hybrid-attention [34], which integrates SA and cross-attention for modality fusion, and then applies the proposed AEIP to perform spatial-influence-guided message passing, where interactions are adaptively emphasized among objects with overlapping influence regions while weak long-range dependencies are suppressed; and 3) an *RSCA-based relation decoder* that injects explicit subject–object relative geometry into relation representations and predicts the final predicates. Overall, SARNet performs relation reasoning in a progressively constrained manner, from global context to instance-centered influence regions, and further to spatial structures within each relation. Such a coarse-to-fine design effectively suppresses noisy long-range interactions, which frequently occur in large-scale remote sensing images.

Given an input image I, the oriented detector produces object features $\boldsymbol{O} \in \mathbb{R}^{N\times d}$, OBBs $\boldsymbol{B}_o \in \mathbb{R}^{N\times 5}$ parameterized by center coordinates, width, height, and orientation, and initial object classification scores $\boldsymbol{L}_o^{\text{init}} \in \mathbb{R}^{N\times C}$

$$\boldsymbol{O}, \boldsymbol{B}_o, \boldsymbol{L}_o^{\text{init}} = \text{OrientedDetector}\,(I) \tag{1}$$

where N denotes the number of objects, d is the dimension of object features and relation features, and C represents the number of object categories.

For all possible object pairs, we apply pair pruning (PPG [1]) to prune the full pair set into a candidate edge set $\boldsymbol{\mathcal{E}}$ containing M relations. For each relation $(i, j) \in \boldsymbol{\mathcal{E}}$ between subject and object, we derive an oriented union bounding box $\boldsymbol{b}_{ij} \in \boldsymbol{B}_r$ that tightly encloses the two objects, and then use it to pool the feature map of the whole image to obtain the relation representation $\boldsymbol{r}_{i\to j} \in \boldsymbol{R}$ for predicate prediction

$$\boldsymbol{R}, \boldsymbol{\mathcal{E}}, \boldsymbol{B}_r = \text{RelationExtract}\left(I, \boldsymbol{B}_o, \boldsymbol{L}_o^{\text{init}}\right) \tag{2}$$

where $\boldsymbol{B}_r \in \mathbb{R}^{M\times 5}$ and $\boldsymbol{R} \in \mathbb{R}^{M\times d}$ denote oriented union boxes and features of all candidate relations, respectively.

The object features are further enhanced by propagating the interaction context with an object enhancer

$$\tilde{\boldsymbol{O}}, \boldsymbol{L}_o^{\text{final}} = \text{ObjectEnhancer}\left(\boldsymbol{O}, \text{Emb}_{sp}\left(\boldsymbol{B}_o\right), \text{Emb}_{se}\left(\boldsymbol{L}_o^{\text{init}}\right)\right) \tag{3}$$

where $\text{Emb}_{sp}(\cdot)$ and $\text{Emb}_{se}(\cdot)$ represent spatial and semantic embeddings, respectively. $\tilde{\boldsymbol{O}}$ denotes the refined object features and $\boldsymbol{L}_o^{\text{final}}$ denotes the updated object classification scores.

For relation prediction, the relation encoder is applied to encode the context-enriched object and relation features

$$\boldsymbol{O}', \boldsymbol{R}' = \text{RelationEncoder}\left(\left[\boldsymbol{O}; \tilde{\boldsymbol{O}}\right], \text{Emb}_{se}\left(\boldsymbol{L}_o^{\text{final}}\right), \boldsymbol{R}\right) \tag{4}$$

where $[\cdot;\cdot]$ denotes concatenation. The relation decoder then predicts the final predicate scores $\boldsymbol{L}_r^{\text{final}}$

$$\boldsymbol{L}_r^{\text{final}} = \text{RelationDecoder}\left(\boldsymbol{O}', \boldsymbol{R}'\right). \tag{5}$$

The relation encoder is built by stacking two blocks, each consisting of a hybrid-attention module followed by AEIP. Within each block, hybrid-attention captures global interactions among all objects to update the object features, while AEIP models an instance-centered local influence region for each object to enhance its awareness of the surrounding environment and update both object and relation features. In

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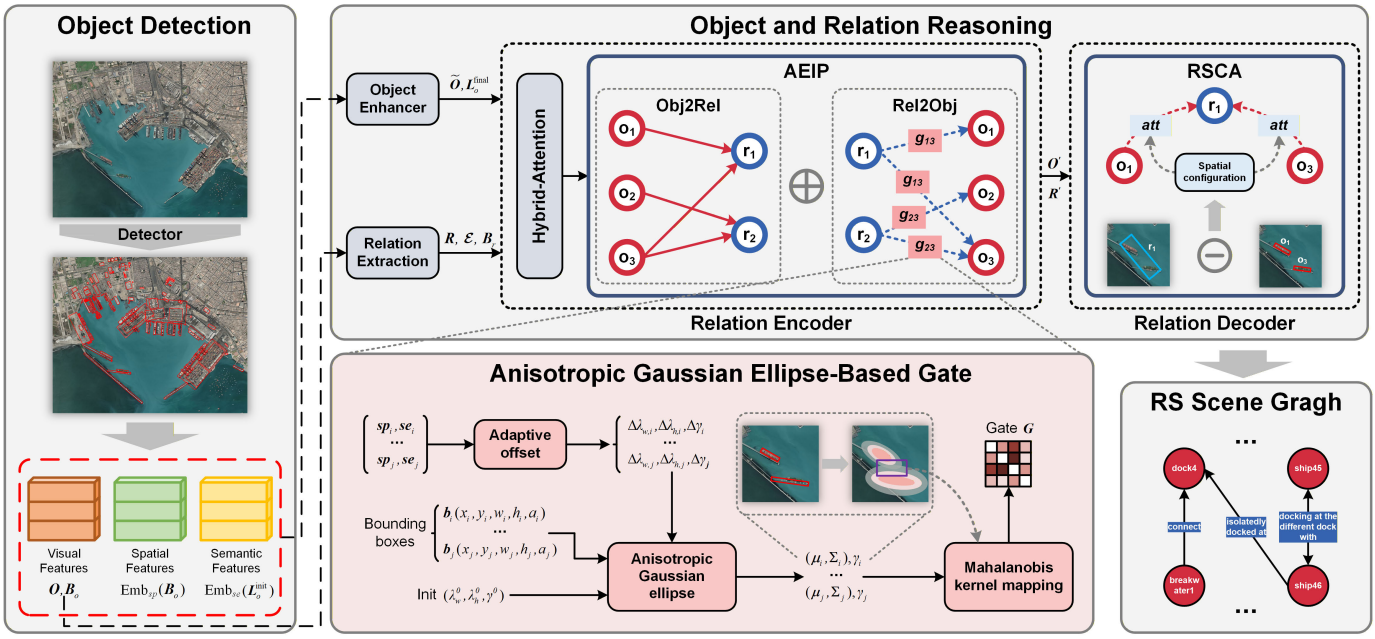


Fig. 2. Overall structure of SARNet. In the first stage, object detection identifies multiscale objects and extracts their features. In the second stage, the enhanced object features and the relation features are processed by the relation encoder consisting of hybrid-attention and AEIP. Hybrid-attention captures global interactions among all objects, while AEIP models an instance-centered local influence region for each object, improving its awareness of the surrounding environment. Next, the relation decoder consisting of four RSCA layers performs subject–object structure-aware reasoning for each relation and predicts the final predicates.

the relation decoder, RSCA performs reasoning over each relation feature by modeling the spatial configuration of the subject and object. Four stacked RSCA layers update the relation features before the final relation classifier. Next, we describe these two modules in detail.

B. Anisotropic Elliptical Influence Propagation

Transformer-based SGG models (e.g., SHA [34] and BGT-Net [33]) perform global attention over all objects. While effective in natural images, such fully global interactions often introduce substantial noise in large-area remote sensing scenes with thousands of instances. MPNNs enable localized reasoning, but they usually apply fixed interaction strengths over neighboring objects, without considering that each object may have its own spatial influence field. In remote sensing imagery, where object interactions vary with distance and spatial configuration, such uniform message passing is insufficient for accurate relation modeling.

To bridge this gap, we propose AEIP for object-centered gated message passing which enables each object to selectively aggregate its relational contextual information from its instance-centered influence region. Specifically, AEIP performs coupled bidirectional updates between object and relation features, where the relation-to-object context flow is controlled by the sparse symmetric gates constructed from OBB-based geometry and predicted semantics, forming instance-centered local influence region perception. In the following parts, we will first introduce the object-centered gated message passing process (see Section III-B1), then detail the construction of the sparse symmetric gates (see Section III-B2).

1) Object-Centered Gated Message Passing: Given object and relation features ($\boldsymbol{O}$ and $\boldsymbol{R}$), we perform bidirectional message passing between object nodes and relation edges, including object-to-relation propagation and relation-to-object propagation. Since the candidate relations are generally much more than objects in remote sensing scenes, aggregating all relation messages can easily overwhelm object representations. Meanwhile, relation features often carry richer surrounding context. We therefore gate the relation-to-object context flow with a sparse symmetric matrix $\boldsymbol{G}$ constructed from the anisotropic Gaussian ellipses defined based on the OBBs, allowing each object to selectively aggregate relation context from its instance-centered local influence region, as shown in the bottom middle part of Fig. 2.

In each AEIP module, both object-to-relation and relation-to-object message passing are performed for two iterations. For the lth iteration, $\boldsymbol{o}_i^l$ and $\boldsymbol{r}_{i\to j}^l$ denote the object (node) and relation (edge), respectively. Specifically,

$$\boldsymbol{r}_{i\to j}^{l+1} = \boldsymbol{r}_{i\to j}^{l} + \phi\left(\boldsymbol{W}_s^{\text{obj}}\boldsymbol{o}_i^l + \boldsymbol{W}_o^{\text{obj}}\boldsymbol{o}_j^l\right) \tag{6}$$

where $\phi(\cdot)$ is an activation function (e.g., ReLU), and $\boldsymbol{W}_s^{\text{obj}}$ and $\boldsymbol{W}_o^{\text{obj}}$ are the projection matrices. The gated relation-to-object message passing process is

$$\boldsymbol{o}_i^{l+1} = \boldsymbol{o}_i^l + \phi\left(\sum_{j\in\mathcal{A}_s(i)} \mathcal{F}_s\left(G_{ij}, \boldsymbol{W}_s^{\text{rel}}\right)\boldsymbol{r}_{i\to j}^l + \sum_{j\in\mathcal{A}_o(i)} \mathcal{F}_o\left(G_{ji}, \boldsymbol{W}_o^{\text{rel}}\right)\boldsymbol{r}_{j\to i}^l\right) \tag{7}$$

where $\boldsymbol{W}_s^{\text{rel}}$ and $\boldsymbol{W}_o^{\text{rel}}$ denote the learnable matrices used to propagate relation messages when node i acts as the *subject*

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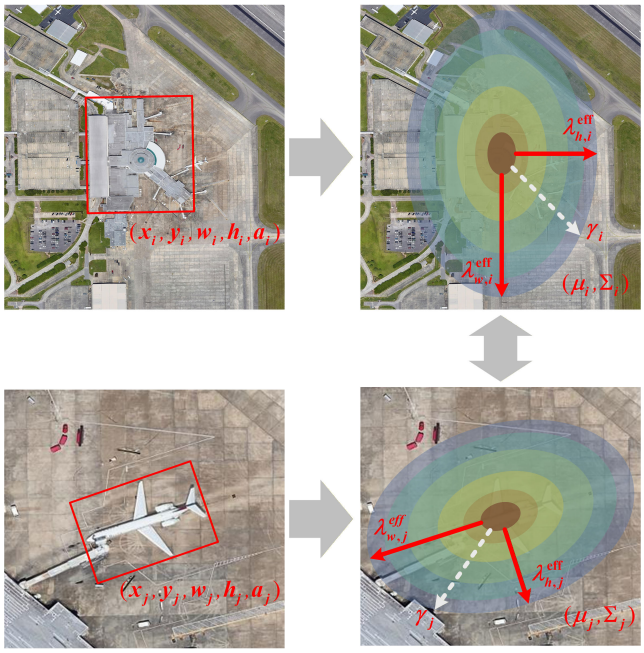


Fig. 3. Illustration of the anisotropic Gaussian ellipse defined based on the OBB.

and *object*, respectively. Accordingly, the mappings $\mathcal{F}_s(\cdot)$ and $\mathcal{F}_o(\cdot)$ perform learnable fusion of the gate $\boldsymbol{G}$ with $\boldsymbol{W}_s^{\text{rel}}$ and $\boldsymbol{W}_o^{\text{rel}}$, respectively. The outgoing and incoming neighbor sets of node i are denoted by $\boldsymbol{\mathcal{A}}_s(i)$ and $\boldsymbol{\mathcal{A}}_o(i)$. $\boldsymbol{G}$ is shared across iterations, whose construction is described in the following parts.

2) Anisotropic Gaussian Ellipse-Based Gate: We construct the gate $\boldsymbol{G} \in \mathbb{R}^{N\times N}$ using the OBB geometry and predicted semantics, which is a sparse and symmetric weight matrix over the candidate edge set $\boldsymbol{\mathcal{E}}$.

As illustrated in Fig. 3, we transform an OBB into a 2-D Gaussian distribution with an ellipse shape, which describes its spatial influence region. The ellipse is aligned with the OBB, where the major and minor axes follow the width and height directions of the box, respectively. In this sense, the ellipse serves as a soft extension of the OBB, providing a continuous influence field whose value gradually decays with distance from the center. This section first introduces the computation of the ellipse parameters, and then describes how the relation weight between two objects (i and j) is computed from their elliptical influence values, resulting in the gating values $\boldsymbol{G}_{ij}$.

a) Anisotropic Gaussian ellipse: Given an OBB $\boldsymbol{b}_i = (x_i, y_i, w_i, h_i, a_i)$, we map it to an anisotropic Gaussian ellipse corresponding to a 2-D Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, where the mean $\boldsymbol{\mu}_i$ is set to the box center, i.e., $\boldsymbol{\mu}_i = [x_i, y_i]^\top$.

Next, in the box-aligned local coordinate system, we construct an axis-aligned diagonal covariance $\boldsymbol{\Lambda}_i$, which defines the ellipse shape. The principal-axis scales are jointly determined by w_i, h_i, and two learnable effective expansion factors along the width and height directions, denoted by $\lambda_{w,i}^{\text{eff}}$ and $\lambda_{h,i}^{\text{eff}}$

$$\boldsymbol{\Lambda}_i = \text{diag}\left(\left(\lambda_{w,i}^{\text{eff}} w_i\right)^2 + \delta,\ \left(\lambda_{h,i}^{\text{eff}} h_i\right)^2 + \delta\right). \tag{8}$$

$\lambda_{w,i}^{\text{eff}}$ and $\lambda_{h,i}^{\text{eff}}$ are generated adaptively on a per-instance basis, and the generation process will be introduced in the next part. These factors control how much the ellipse expands relative to $\boldsymbol{b}_i$, making the ellipse scale strongly coupled with w_i and h_i. The constant δ is a numerical-stability term that enforces a minimum variance floor for the diagonal covariance entries under normalized coordinates.

Finally, we rotate the ellipse defined in the box-aligned local coordinate system into the global image coordinate system. The rotation matrix $\boldsymbol{R}_\theta$ and the resulting global covariance $\boldsymbol{\Sigma}_i$ are

$$\boldsymbol{R}_\theta = \begin{bmatrix} \cos a_i & -\sin a_i \\ \sin a_i & \cos a_i \end{bmatrix} \tag{9}$$

$$\boldsymbol{\Sigma}_i = \boldsymbol{R}_\theta \boldsymbol{\Lambda}_i \boldsymbol{R}_\theta^\top. \tag{10}$$

In this way, each OBB is mapped to a Gaussian distribution $\mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, whose principal axes align with the box orientation and whose scale is adaptively expanded with the box size. This provides an interpretable geometric prior for the subsequent ellipse-distance-based relation-to-object propagation gating.

b) Adaptive offset-based expansion factors: As defined previously, the effective expansion factors $\lambda_{w,i}^{\text{eff}}$ and $\lambda_{h,i}^{\text{eff}}$ are introduced to control the scaling of the elliptical influence region of instance i along the width and height axes. Besides, to model the distance-dependent decay rate of the influence weights within the elliptical region, we introduce another effective distance-aware decay strength factor γ_i. We define the computation of these factors as follows:

$$\begin{aligned} \lambda_{w,i}^{\text{eff}} &= \mathcal{S}\left(\lambda_w^0 + \Delta\lambda_{w,i}\right) \\ \lambda_{h,i}^{\text{eff}} &= \mathcal{S}\left(\lambda_h^0 + \Delta\lambda_{h,i}\right) \\ \gamma_i &= \text{Softplus}\left(\gamma^0 + \Delta\gamma_i\right) \end{aligned} \tag{11}$$

where λ_w^0, λ_h^0, and γ^0 are the initial constant parameters. $\mathcal{S}(\cdot)$ denotes a bounded mapping constructed from the sigmoid function, which constrains the effective expansion factors to the predefined range $(0.05, 5.0)$, whereas the Softplus operation ensures that the decay factor γ_i remains strictly positive. These parameterizations prevent excessively small or large influence regions and promote stable gate computation. The instance-wise adaptive offsets $\Delta\lambda_{w,i}$, $\Delta\lambda_{h,i}$, and $\Delta\gamma_i$ are deliberately designed based on the intuition that an object's influence region is largely governed by its geometric characteristics and semantic category. Concretely, for each object i, we first encode the geometry of its OBB $\boldsymbol{b}_i$ with a two-layer MLP, and compute its semantic feature as a weighted sum of the pretrained GloVe [56] embeddings, resulting in the geometry embedding $\boldsymbol{sp}_i$ and the semantic feature $\boldsymbol{se}_i$

$$\begin{aligned} \boldsymbol{sp}_i &= \text{MLP}\left(\text{Emb}_{sp}\left(\boldsymbol{b}_i\right)\right) \\ \boldsymbol{se}_i &= \text{Softmax}\left(\boldsymbol{\ell}_{o,i}^{\text{final}}\right) \boldsymbol{E}_{se} \end{aligned} \tag{12}$$

where $\boldsymbol{\ell}_{o,i}^{\text{final}}$ is the classification scores for object i. However, in remote sensing imagery, variations in ground sampling distance (GSD) often cause significant scale differences across images, making the geometry and semantics of a single instance insufficient to determine a proper influence range. SA enables each instance to aggregate global contextual cues, such as overall scale and object density, from all other instances in

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the image. Therefore, we predict the 3-D adaptive offset vector $[\Delta\lambda_{w,i}, \Delta\lambda_{h,i}, \Delta\gamma_i]$ based on the resulting geometry embedding and the semantic feature by applying a SA layer and then a fully connected (FC) layer

$$\left[\Delta\lambda_{w,i}, \Delta\lambda_{h,i}, \Delta\gamma_i\right] = \mathrm{FC}\left(\mathrm{SA}\left(\left[\boldsymbol{sp}_i; \boldsymbol{se}_i\right]\right)\right). \tag{13}$$

c) Mahalanobis distance-induced gate weights: After obtaining the Gaussian representation of each object, $\mathcal{N}(\boldsymbol{\mu}_i, \boldsymbol{\Sigma}_i)$, together with the instance-wise adaptive parameters $\lambda_{w,i}^{\mathrm{eff}}$, $\lambda_{h,i}^{\mathrm{eff}}$, and γ_i, we compute the gate weights over the candidate relation set $\boldsymbol{\mathcal{E}}$. For a relation $(i, j) \in \boldsymbol{\mathcal{E}}$, we aggregate the uncertainties of the two instances to form a pairwise covariance $\boldsymbol{\Sigma}_{ij} = \boldsymbol{\Sigma}_i + \boldsymbol{\Sigma}_j$, and define the Mahalanobis distance between the two centers under this ellipse metric. In practice, we introduce an additional scale factor τ to adjust the distance magnitude

$$d_{ij}^2 = \tau\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\right)^\top \boldsymbol{\Sigma}_{ij}^{-1}\left(\boldsymbol{\mu}_i - \boldsymbol{\mu}_j\right). \tag{14}$$

To capture the distance-dependent attenuation of interactions, for each relation $(i, j) \in \boldsymbol{\mathcal{E}}$, we define a symmetric attenuation term γ_{ij}, computed via geometric averaging, to control the attenuation strength of their mutual influence

$$\gamma_{ij} = \sqrt{\gamma_i \gamma_j}. \tag{15}$$

We then incorporate γ_{ij} into the energy term $z_{ij} = \gamma_{ij} d_{ij}^2 > 0$. For numerical stability and smooth gradients, we apply a monotonically increasing SoftCap function to z_{ij}

$$\mathrm{SoftCap}\left(z_{ij}\right) = z_{ij} - \mathrm{Softplus}\left(z_{ij} - z_{\max}\right) \tag{16}$$

where $z_{\max}$ specifies the upper asymptote of the SoftCap output. Accordingly, the output satisfies $\forall\, i, j,\ 0 < \mathrm{SoftCap}(z_{ij}) < z_{\max}$, and is then mapped to the gate weight through the following bounded attenuation function:

$$g_{ij} = \frac{1}{1 + \mathrm{SoftCap}\left(z_{ij}\right)}. \tag{17}$$

Thus, the interaction strength $g_{ij} \in (0, 1)$ decays monotonically with the increasing pairwise distance d_{ij}, with γ_{ij} controlling the decay rate.

The pairwise gate weights g_{ij} are then organized into a gating matrix $\boldsymbol{G} \in \mathbb{R}^{N \times N}$ whose entries are defined only on candidate edges, while symmetry is explicitly enforced and diagonal elements are set to zero. This gating mechanism adaptively modulates the interaction strength between two objects. g_{ij} becomes larger when the two instances exhibit stronger mutual influence, which is governed by their anisotropic ellipse expansions and decay strengths, namely, a larger influence field and slower decay yield stronger interactions under the ellipse metric. Conversely, weak influence leads to a smaller g_{ij}, attenuating the corresponding relation message. In this way, $\boldsymbol{G}$ provides an interpretable geometric prior for relation-to-object context flow.

C. Relative Spatial Configuration-Guided Attention

Relations in remote sensing scenes are closely related to the geometric properties of subjects and objects, particularly their relative spatial configurations. However, existing methods do not explicitly exploit such relative geometric cues during relation reasoning. To address this limitation, we incorporate subject–object relative spatial configuration into the relation decoding stage, enabling fine-grained geometry-aware reasoning and providing a more stable spatial basis for predicate prediction.

Concretely, we introduce the RSCA module to decode spatial relations. For each object pair, RSCA takes the relation feature as the query and performs geometry-modulated cross-attention over its corresponding subject and object features. This design encodes the relative geometric layout of the subject and object with respect to the union region into the relation representation, enabling the model to capture more discriminative spatial cues and reduce ambiguity in predicate prediction.

As shown in Fig. 4, RSCA mainly comprises a geometry-guided attention layer and directly takes the object features $\boldsymbol{O}'$ and relation features $\boldsymbol{R}'$ produced by the relation encoder as inputs. For each relation $(i, j) \in \boldsymbol{\mathcal{E}}$ between the subject i and object j, we embed the OBBs of the subject and object $\boldsymbol{b}_i, \boldsymbol{b}_j \in \boldsymbol{B}_o$, as well as their union OBB $\boldsymbol{b}_{ij} \in \boldsymbol{B}_r$, to obtain the geometry embeddings $\boldsymbol{sp}_i, \boldsymbol{sp}_j$ and $\boldsymbol{sp}_{i\to j}$ for the subject, object, and relation, respectively,

$$\begin{aligned}
\boldsymbol{sp}_i &= \mathrm{MLP}_{node}\left(\mathrm{Emb}_{sp}\left(\boldsymbol{b}_i\right)\right) \\
\boldsymbol{sp}_j &= \mathrm{MLP}_{node}\left(\mathrm{Emb}_{sp}\left(\boldsymbol{b}_j\right)\right) \\
\boldsymbol{sp}_{i\to j} &= \mathrm{MLP}_{edge}\left(\mathrm{Emb}_{sp}\left(\boldsymbol{b}_{ij}\right)\right)
\end{aligned} \tag{18}$$

where $\mathrm{MLP}_{node}(\cdot)$ and $\mathrm{MLP}_{edge}(\cdot)$ are two-layer MLPs.

To explicitly characterize the spatial configurations of the subject and object within the relation region, we construct two relative geometry representations, $\Delta\boldsymbol{sp}_{i\to j}^{i}$ and $\Delta\boldsymbol{sp}_{i\to j}^{j}$, which describe the relative geometric configuration of the subject and object with respect to the union region

$$\begin{aligned}
\Delta\boldsymbol{sp}_{i\to j}^{i} &= \mathrm{MLP}_{geo}\left(\boldsymbol{sp}_{i\to j} - \boldsymbol{sp}_i\right) \\
\Delta\boldsymbol{sp}_{i\to j}^{j} &= \mathrm{MLP}_{geo}\left(\boldsymbol{sp}_{i\to j} - \boldsymbol{sp}_j\right)
\end{aligned} \tag{19}$$

where $\mathrm{MLP}_{geo}(\cdot)$ is a two-layer MLP.

To decode the relation, we form the query from the relation feature $\boldsymbol{r}'_{i\to j} \in \boldsymbol{R}'$, while the key and value are both derived from the corresponding subject/object features $\boldsymbol{o}'_i, \boldsymbol{o}'_j \in \boldsymbol{O}'$

$$\begin{aligned}
\boldsymbol{q}_{i\to j} &= \boldsymbol{W}_q \boldsymbol{r}'_{i\to j} \\
\boldsymbol{k}_{i\to j}^{i} &= \boldsymbol{W}_k \boldsymbol{o}'_i, \quad \boldsymbol{k}_{i\to j}^{j} = \boldsymbol{W}_k \boldsymbol{o}'_j \\
\boldsymbol{v}_{i\to j}^{i} &= \boldsymbol{W}_v \boldsymbol{o}'_i, \quad \boldsymbol{v}_{i\to j}^{j} = \boldsymbol{W}_v \boldsymbol{o}'_j
\end{aligned} \tag{20}$$

where $\boldsymbol{W}_q, \boldsymbol{W}_k$, and $\boldsymbol{W}_v$ are learnable projection matrices.

We compute the geometry-modulated attention weights

$$\boldsymbol{\alpha}_{i\to j}^{n} = \mathrm{Softmax}\left(\frac{\mathrm{MLP}_{att}\left(\boldsymbol{q}_{i\to j} - \boldsymbol{k}_{i\to j}^{n} + \Delta\boldsymbol{sp}_{i\to j}^{n}\right)}{\sqrt{d}}\right) \tag{21}$$

where $n \in \{i, j\}$ and $\mathrm{MLP}_{att}(\cdot)$ is a two-layer MLP. This process restricts attention competition within the subject and object, respectively, reducing interference from irrelevant instances in dense remote sensing scenes. The relative geometry encodings $\Delta\boldsymbol{sp}_{i\to j}^{i}$ and $\Delta\boldsymbol{sp}_{i\to j}^{j}$ provide relation-aware spatial cues that modulate the attention to the subject and object features,

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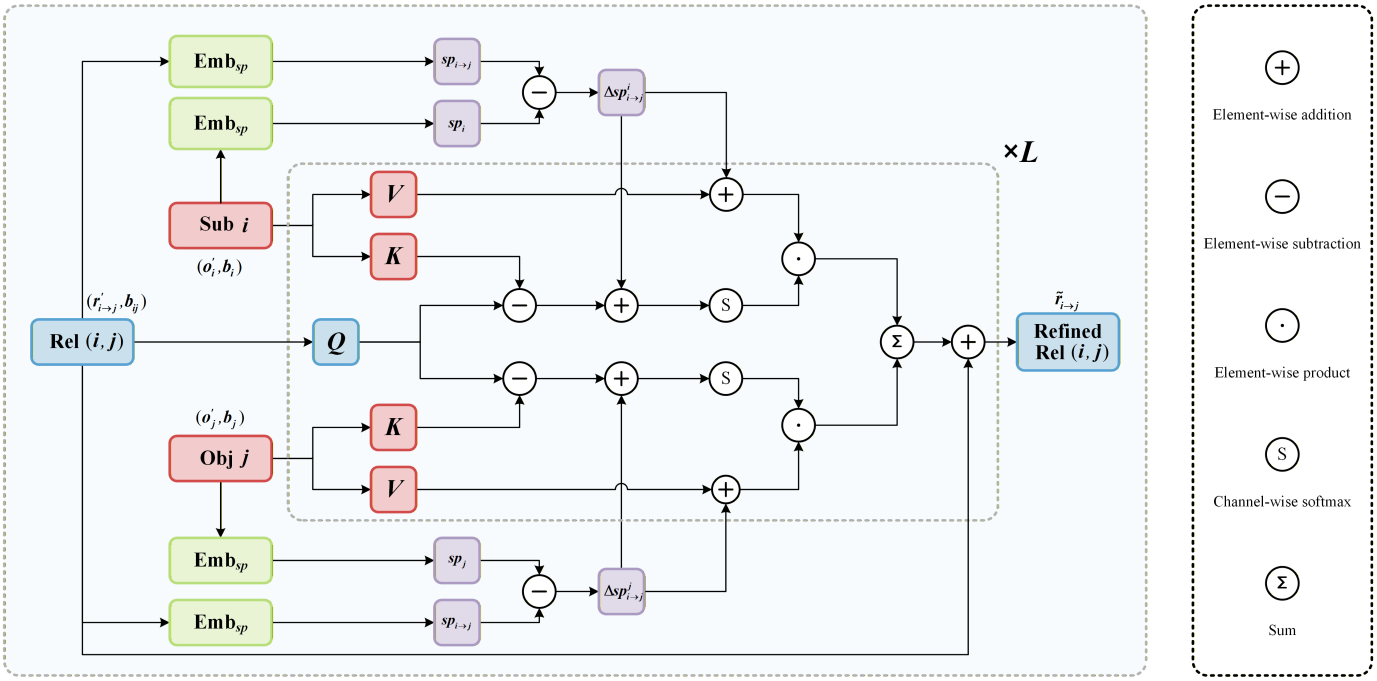


Fig. 4. Structure of RSCA. The relation feature serves as the query for geometry-aware differential matching and cross-attention aggregation over the subject and object features, incorporating object information with relative subject–object spatial configurations into the relation feature.

thereby embedding their spatial configuration within the union region into the refined relation feature. Based on the resulting dispatch weights, we aggregate geometry-enhanced messages as follows:

$$\boldsymbol{m}_{i\to j} = \sum_{n\in\{i,j\}} \boldsymbol{\alpha}^n_{i\to j} \odot \left(\boldsymbol{v}^n_{i\to j} + \Delta \boldsymbol{sp}^n_{i\to j}\right) \tag{22}$$

which is then passed through an FC layer FC(·), and the output is added back to the relation feature for residual refinement

$$\widetilde{\boldsymbol{r}}_{i\to j} = \boldsymbol{r}'_{i\to j} + \mathrm{FC}\left(\boldsymbol{m}_{i\to j}\right) \tag{23}$$

$\widetilde{\boldsymbol{r}}_{i\to j}$ preserves the original relation semantics and explicitly incorporates the relative spatial configuration of the subject and object with respect to the union region. This makes predicate inference more sensitive to discriminative geometric cues and reduces ambiguity.

In our implementation, we stack $L = 4$ RSCA layers and each layer takes the same initial relation feature as input. The final representation is obtained by averaging the outputs of all layers

$$\boldsymbol{r}^{\text{final}}_{i\to j} = \frac{1}{L}\sum_{l=1}^{L} \widetilde{\boldsymbol{r}}^l_{i\to j}. \tag{24}$$

After obtaining the final relation representation $\boldsymbol{r}^{\text{final}}_{i\to j}$, the subject and object features are concatenated to construct the object-pair feature, which is projected and then fused with the relation representation through element-wise multiplication. The fused representation is fed into a classifier to produce the final predicted relation scores $\boldsymbol{l}^{\text{final}}_{r,i\to j}$

$$\boldsymbol{l}^{\text{final}}_{r,i\to j} = \mathrm{FC}\left(\mathrm{FC}\left(\left[\boldsymbol{o}'_i; \boldsymbol{o}'_j\right]\right) \odot \boldsymbol{r}^{\text{final}}_{i\to j}\right). \tag{25}$$

D. Loss Function

We jointly optimize object classification and predicate classification. The overall loss function consists of an object classification loss $\mathcal{L}_{obj}$ and a predicate classification loss $\mathcal{L}_{rel}$

$$\mathcal{L} = \mathcal{L}_{obj} + \mathcal{L}_{rel}. \tag{26}$$

The $\mathcal{L}_{obj}$ and $\mathcal{L}_{rel}$ are cross-entropy losses as follows:

$$\mathcal{L}_{obj} = -\frac{1}{N}\sum_{i=1}^{N} \log\left(\mathrm{Softmax}\left(\boldsymbol{l}^{\text{final}}_{o,i}\right)\left(y_{o,i}\right)\right) \tag{27}$$

$$\mathcal{L}_{rel} = -\frac{1}{M}\sum_{(i,j)\in\boldsymbol{\mathcal{E}}} \log\left(\mathrm{Softmax}\left(\boldsymbol{l}^{\text{final}}_{r,i\to j}\right)\left(y_{r,i\to j}\right)\right) \tag{28}$$

where $\boldsymbol{l}^{\text{final}}_{o,i}$, $\boldsymbol{l}^{\text{final}}_{r,i\to j}$ represent the predicted object and relation scores, respectively, and $y_{o,i}$ and $y_{r,i\to j}$ denote the corresponding ground-truth (GT) labels. The terms Softmax$(\boldsymbol{l}^{\text{final}}_{o,i})(y_{o,i})$ and Softmax$(\boldsymbol{l}^{\text{final}}_{r,i\to j})(y_{r,i\to j})$ denote the predicted scores of the GT object category and predicate, respectively.

IV. Experiments

In this section, we evaluate the effectiveness of the proposed SARNet on STAR [1] and AUG [28], two widely adopted benchmarks. The dataset information, evaluation tasks and metrics, and the implementation details are introduced in Sections IV-A–IV-D, respectively. Subsequently, we provide a performance comparison with state-of-the-art methods in Section IV-E. To quantify the contribution of every component in SARNet, a comprehensive ablation analysis is provided in Section IV-F. We perform some further explorations in Sections IV-G–IV-I.

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A. Dataset

STAR [1] is the first large-scale RS-SGG dataset tailored to large-area remote sensing images. It contains over 1200 representative scenes worldwide, covering transportation, energy, and daily-life activities. The image sizes range from 512×768 to $27\,860 \times 31\,096$. STAR contains 48 object categories and 58 relation categories, covering more than 210k annotated objects and 400k annotated triplets. Following the official protocol [1], we split the dataset into training/validation/test sets with a 60%/20%/20% ratio. Due to the extremely large spatial coverage and scene diversity, a single image may include up to 6888 objects, which poses a significant challenge for modeling global object interactions. Moreover, STAR exhibits substantial intraclass variations and a pronounced long-tailed distribution over relation categories.

AUG [28] is an aerial image urban SGG dataset extended from the Semantic Drone Dataset, with images captured by UAVs at approximately 5–35 m above the ground. AUG contains 400 high-resolution urban aerial images with a resolution of 6000×4000. It is manually annotated with 25 594 objects and 16 970 triplets. Our experiments strictly follow the dataset split adopted in the original AUG study [28], with 360 images used for training and 38 images for testing.

B. Evaluation Tasks

We evaluate SARNet on three standard SGG and RS-SGG tasks as follows.

1) *Predicate Classification (PredCls):* Given GT OBBs and object categories, the model only predicts the predicate category for each object pair. This setting eliminates interference from object detection errors and directly evaluates relation prediction capability.
2) *Scene Graph Classification (SGCls):* Given GT OBBs, the model predicts both object categories and predicates. This setting evaluates joint object recognition and relation understanding.
3) *Scene Graph Detection (SGDet):* Given only an input image, the model first detects objects (OBBs and categories) and then predicts predicates. A predicted box is considered correct if its IoU with the GT box is over 0.5. SGDet is the most challenging and realistic setting, as detection noise can significantly degrade relation inference.

C. Evaluation Metrics

For the STAR dataset, following its evaluation protocol, we adopt multilabel recall@K (MR@K) and mean multilabel recall@K (mMR@K), with $\mathrm{K} \in \{1500, 2000\}$. For the AUG dataset, we report recall@K (R@K) and mean recall@K (mR@K), with $\mathrm{K} \in \{50, 150\}$.

MR@K computes the recall rate of the correctly predicted top-K subject-predicate-object triplets. As each unique subject–object pair is annotated with multiple predicates, the predicted triplet is correct as long as its predicate matches one of the annotated predicates. It can be written as

$$\mathrm{MR@K} = \frac{\mathrm{TP}_{pair}\mathrm{@K}}{\left|\mathrm{UniquePairs\,(GT)}\right|} \tag{29}$$

where TP_{pair}@K denotes the number of correctly retrieved *unique* (subject, object) pairs, and UniquePairs(GT) is the set of unique (subject and object) pairs that have at least one annotated relation. R@K measures the fraction of GT triplets that are correctly retrieved among the top-K predictions

$$\mathrm{R@K} = \frac{\mathrm{TP@K}}{|\mathrm{GT}|} \tag{30}$$

where TP@K denotes the number of correctly predicted subject-predicate-object triplets within the top-K predictions and |GT| is the number of GT triplets.

Due to the long-tailed relation distribution, recall-based metrics may overemphasize frequent predicates. Therefore, mMR@K and mR@K further average the corresponding recall over relation categories

$$\mathrm{mMR@K} = \frac{1}{C}\sum_{c=1}^{C}\mathrm{MR}_c\mathrm{@K} \tag{31}$$

$$\mathrm{mR@K} = \frac{1}{C}\sum_{c=1}^{C}\mathrm{R}_c\mathrm{@K} \tag{32}$$

where C is the number of predicate categories, with MR_c@K and R_c@K for category c on STAR and AUG, respectively.

D. Implementation Details

Following the standard two-stage setting, we freeze the pretrained object detectors and train only the SGG models. Across the two datasets, we use SGD with a momentum of 0.9 and a weight decay of 1×10^{-4}, together with a linear warm-up over the first 500 iterations, starting at 0.1 times the base learning rate. The checkpoint at the final predefined training iteration is used for evaluation in all tasks. The AEIP parameters τ, $z_{\max}$, and δ are consistently set to 0.5, 40, and 1×10^{-8}, respectively. All experiments are conducted on NVIDIA L20 GPUs.

For STAR, we use HOD-Net [1] as the pretrained oriented object detector and PPG [1] for relation-pair pruning during inference. The random seed is fixed to 1029. PredCls and SGCls are trained for 10 000 iterations with a batch size of 4, whereas SGDet is trained for 5000 iterations with a batch size of 2. The base learning rate is 1×10^{-3} and is reduced by a factor of 0.1 at 6000 and 8500 iterations for PredCls and SGCls, and at 3000 and 4000 iterations for SGDet.

For AUG, we use Faster R-CNN [57] with a VGG16 [58] backbone as the pretrained detector. For a fair comparison, we reproduce all competing methods using the same detector and training protocol. Models for all three tasks are trained for 40 000 iterations with a batch size of 2 and an initial learning rate of 5×10^{-3}. Following warm-up, performance is evaluated every 2000 iterations. The learning rate is reduced by a factor of 0.5 after six consecutive evaluations without improvement, followed by a cooldown of two evaluation periods.

E. Performance Comparison

To comprehensively assess relation reasoning in complex remote sensing scenes, we compare SARNet with representative state-of-the-art NI-SGG and RS-SGG methods on STAR

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TABLE I
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON STAR. † DENOTES THE USE OF SEMANTIC FILTERING FOR SGDET. NI-SGG DENOTES NATURAL-IMAGE SGG

Method	PredCls				SGCls				SGDet			
	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000
NI-SGG												
IMP[20]	51.52	53.19	21.47	22.31	46.08	47.50	18.57	19.38	18.13	19.03	5.60	5.99
Motifs[25]	62.02	64.00	29.40	30.64	50.44	52.29	24.21	25.36	20.63	21.60	7.78	8.22
GPSNet[30]	65.38	66.87	32.52	34.08	29.17	30.36	14.28	14.90	24.18	25.04	9.54	10.09
SHA[34]	65.00	67.04	33.26	34.78	51.01	52.63	23.30	24.15	28.01	29.06	10.18	10.79
HETSGG[32]	63.81	65.37	29.74	30.98	27.64	28.96	12.18	12.78	19.45	20.32	5.44	5.73
PENet[59]	62.87	64.98	36.99	38.29	41.79	43.45	20.64	21.60	21.26	22.50	8.75	9.30
RS-SGG												
HMPNet[45]	62.90	65.10	37.50	38.60	48.30	49.70	25.70	26.10	25.50	26.30	10.20	11.00
SSMPNet[48]	65.30	66.70	**42.50**	**43.90**	52.70	53.90	29.30	30.40	28.20	29.60	11.20	12.30
RPCM[1]	64.23	65.86	41.24	42.30	51.29	52.72	**30.04**	**30.85**	27.23	28.50	11.53	12.07
SARNet	**69.06**	**70.77**	39.53	40.67	**54.41**	**55.65**	27.91	28.73	**33.52**	**34.56**	**14.05**	**14.61**
RPCM†[1]	–	–	–	–	–	–	–	–	30.86	31.72	13.62	14.05
SARNet†	–	–	–	–	–	–	–	–	**35.73**	**36.64**	**14.88**	**15.37**

and report results across all three tasks. To further evaluate the generalization capability of SARNet, we conduct additional comparative experiments on the AUG dataset. AUG consists of UAV overhead-view images annotated with horizontal bounding boxes (HBBs), providing a complementary setting that differs from typical NI-SGG while being less geographically large-scale than standard remote sensing SGG benchmarks like STAR.

Notably, RPCM [1] employs an additional semantic filtering step during inference for PredCls and SGCls, but not for SGDet. For a fair comparison, we report the results of both RPCM and SARNet with and without semantic filtering; the results with filtering are marked by †. For all subsequent experiments on STAR following Table I, including the ablation studies, the same semantic filtering protocol is consistently applied to all three tasks, and it is not applied in the experiments on AUG.

1) Quantitative Comparison:

a) Performances on STAR: From Table I, it can be observed that RS-SGG methods consistently outperform NI-SGG methods, suggesting that the unique characteristics of remote sensing images prevent NI-SGG models from achieving their full potential.

When comparing SARNet with RS-SGG methods, the following observations can be made.

PredCls: SARNet achieves the best MR, reaching 69.06%/70.77% on MR@1500/MR@2000. It improves over the previous best method by 3.76%/3.73% on MR@1500/MR@2000. For mMR@1500/mMR@2000, SARNet is slightly lower than the top-performing methods (e.g., SSMPNet and RPCM), while still clearly outperforming other methods.

SGCls: SARNet achieves 54.41%/55.65% on MR@1500/MR@2000, surpassing the SOTA method SSMPNet by 1.71%/1.75%. Its mMR@1500/mMR@2000 is slightly below the top two methods, but remains substantially higher than other competitors.

SGDet: Under the most challenging setting, SARNet significantly surpasses previous methods, achieving 33.52%/34.56% on MR@1500/MR@2000. Compared with a recent SOTA method SSMPNet, SARNet improves MR@1500/MR@2000 by 5.32%/4.96%. Moreover, SARNet achieves 14.05%/14.61% on mMR@1500/mMR@2000, surpassing the best and second-best methods by 2.52%/2.31%. These results indicate that SARNet achieves the largest gains on SGDet, reflecting its inherent robustness against severe detection noise. After applying semantic filtering, SARNet significantly outperforms a recent SOTA method RPCM on SGDet. Specifically, SARNet achieves 35.73%/36.64% on MR@1500/MR@2000 and 14.88%/15.37% on mMR@1500/mMR@2000.

Analysis of Relation Prediction: Overall, SARNet consistently achieves higher MR across all tasks while remaining competitive on mMR. Compared with natural-image benchmarks, remote sensing typically involves fewer relation categories (e.g., 58 in STAR versus 150 in VG [60]), but far more objects and relations per image. Therefore, when assessing remote sensing scene graph quality, recall metrics (MR@K) deserve particular attention, in addition to category-balanced metrics (mMR@K).

We further compare the performance of RPCM and SARNet in terms of per-predicate prediction accuracy. Fig. 5 shows the MR@K values of RPCM and SARNet for each predicate category on the SGDet task, where predicates are ordered from left to right by descending frequency. On the most challenging SGDet task, SARNet correctly predicts predicates that require both awareness of each object's surrounding influence context and precise perception of spatial distributions among objects, such as "parked alongside with," "in the different parking with," "docking at the different dock with," and "within different line of." It is noted that we do not apply additional debiasing strategies. RPCM struggles to recognize these predicates. This further highlights the importance of the perception of the object-centered local influence region.

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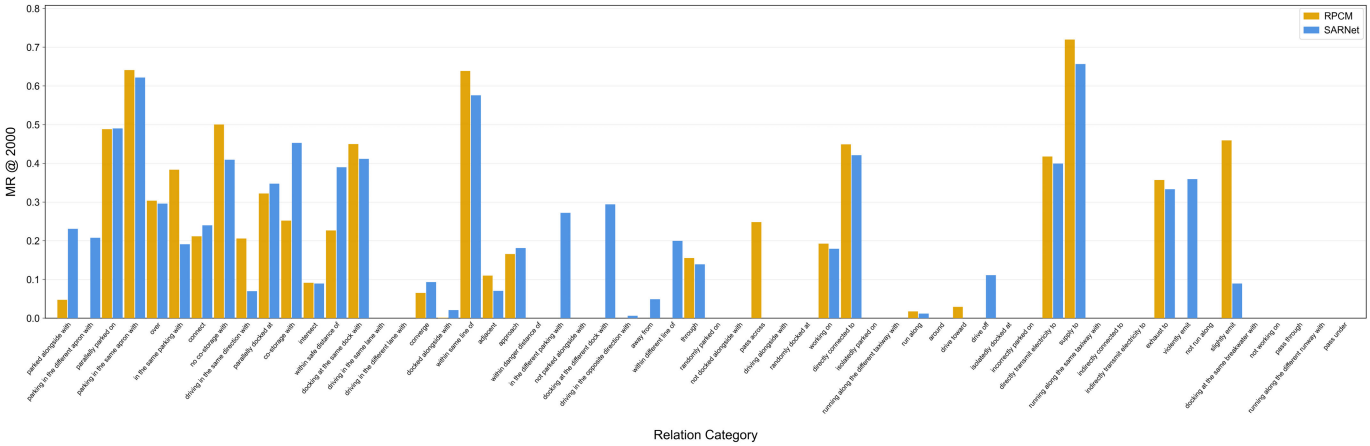


Fig. 5. MR@2000 performance comparison of RPCM and SARNet for each predicate on the SGDet task. Predicates are sorted in descending order from left to right according to the frequency.

TABLE II
PERFORMANCE COMPARISON OF DIFFERENT METHODS ON AUG

Method	PredCls				SGCls				SGDet			
	R@50	R@150	mR@50	mR@150	R@50	R@150	mR@50	mR@150	R@50	R@150	mR@50	mR@150
Motifs[25]	24.65	31.81	9.87	12.85	21.14	26.57	6.34	8.65	15.28	18.31	3.99	4.89
VCTree[61]	28.15	38.00	14.41	18.19	18.70	23.55	3.96	5.11	13.62	19.63	3.06	4.48
SQUAT[36]	24.93	34.47	8.67	13.33	17.31	23.84	3.37	6.54	12.73	18.56	3.55	6.02
PE-Net[59]	**32.41**	**41.94**	**16.26**	20.49	22.60	27.08	6.75	8.45	**15.94**	20.09	4.55	5.80
RPCM[1]	30.80	41.63	16.14	**20.96**	22.52	26.97	7.13	9.11	13.57	19.13	**5.04**	6.75
SARNet	28.85	39.19	13.34	17.69	**23.82**	**30.54**	**7.47**	**9.81**	15.43	**22.20**	4.75	**8.02**

On the PredCls task, where object boxes and labels are given, the strong MR and mMR scores demonstrate SARNet's effective predicate reasoning. On the SGCls task, moderate noise is introduced by object classification errors; SARNet remains competitive but is relatively less dominant, which is consistent with the fact that SARNet does not introduce additional dedicated designs specifically for improving object classification. On the SGDet task, detection errors introduce substantial noise. SARNet achieves the largest performance gains under this setting, which is consistent with our design that progressively decouples reasoning from global context to object-centered anisotropic influence context, and further to intrarelation spatial structures. This coarse-to-fine reasoning scheme retains global awareness while suppressing noisy long-range interactions, leading to more robust relation inference in large-area remote sensing scenes.

b) Performances on AUG: The comparison results on the AUG dataset are reported in Table II. On PredCls, SARNet achieves 28.85%/39.19% on R@50/R@150 and 13.34%/17.69% on mR@50/mR@150, showing competitive performance. When object classification or detection noise is introduced, SARNet exhibits clearer advantages. On SGCls, it achieves the best performance across all metrics, reaching 23.82%/30.54% on R@50/R@150 and 7.47%/9.81% on mR@50/mR@150, improving over the previous best methods by 1.22%/3.46% and 0.34%/0.70%, respectively. On SGDet, SARNet obtains the best R@150 and mR@150 scores, surpassing the previous best results by 2.11% and 1.27%, respectively.

Compared with STAR, AUG uses HBB annotations and is collected from UAV overhead views with more localized scene layouts, which may weaken the advantages of orientation-aware anisotropic influence modeling in AEIP and relative spatial configuration perception in RSCA. Specifically, AEIP is designed to exploit orientation-aware object geometry derived from OBBs, while HBB annotations provide less explicit directional information for modeling anisotropic spatial influence. Nevertheless, SARNet still achieves competitive performance on AUG, especially under SGCls and SGDet, where object recognition and detection noise are introduced. The best or second-best results on both R and mR metrics in these settings indicate that SARNet retains favorable generalization ability beyond standard OBB-based remote sensing SGG benchmarks. The proposed progressive spatial reasoning design can generalize to HBB-based UAV scenes and provide robustness against recognition and detection uncertainties, even though the suppression of large-scale noisy interactions is less explicitly reflected than on STAR.

2) Qualitative Analysis: The RS-SGG results of RPCM and SARNet in static scenes (e.g., airports and ports) and dynamic scenes (e.g., highways and waterways) are shown in Figs. 6 and 7, respectively. For predicates, blue, red, purple, and gray edges denote correct predictions, incorrect predictions, reasonable but unannotated predictions, and missing predictions, respectively. Their results in large-extent scenes (e.g., multidock ports and intersecting-runway airports) are illustrated in Fig. 8. SARNet correctly predicts relations that require both object-centered influence context understanding

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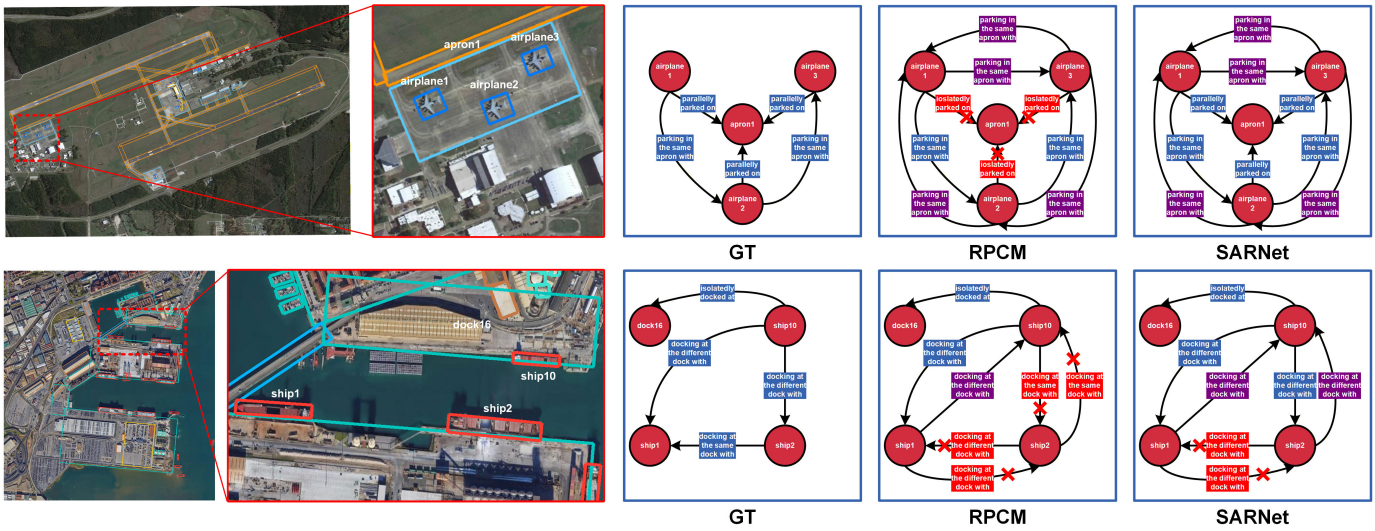


Fig. 6. Visualization results of RPCM and SARNet on static scenes from the STAR test set. For predicates, blue edges denote correct predictions, red edges represent incorrect predictions, and purple edges indicate reasonable predictions but not annotated in the GT.

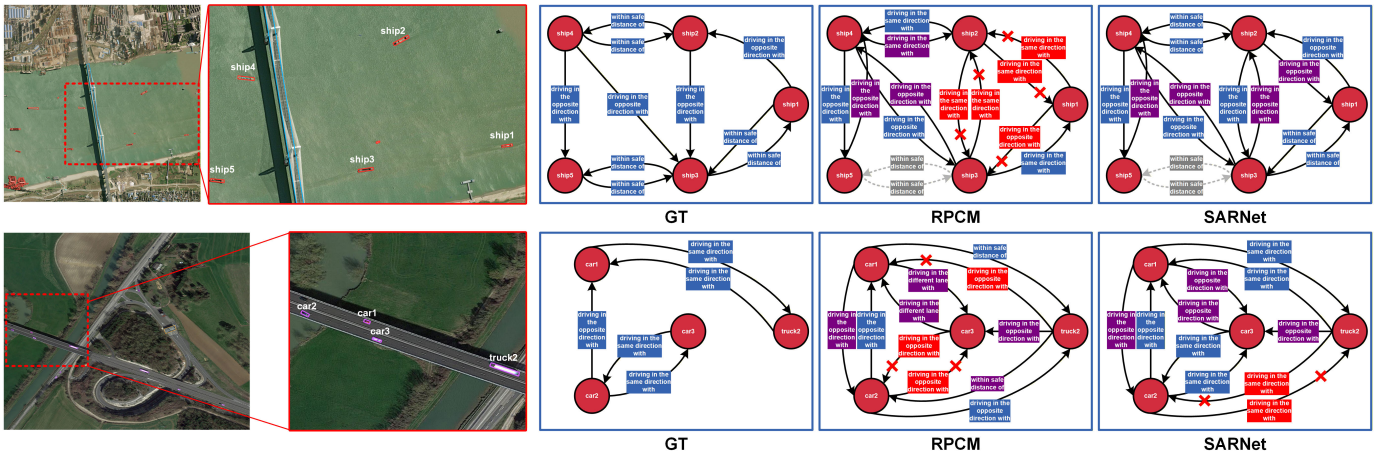


Fig. 7. Visualization results of RPCM and SARNet on dynamic scenes from the STAR test set. Gray edges represent missing predicates. The other edge colors have the same meanings as those in Fig. 6.

and perception of object spatial distributions. In static scenes, as shown in the top of Fig. 6, SARNet allows airplanes within the same apron to perceive each other, and thus correctly predicts ⟨*airplane0/airplane1/airplane2, parallelly parked on, apron16*⟩. By contrast, methods with limited emphasis on each object's local context may incorrectly produce ⟨*airplane0/airplane1/airplane2, isolatedly parked on, apron16*⟩. In dynamic scenes, as shown in the top example of Fig. 7, SARNet correctly captures the relative navigational states of ships, such as ⟨*ship1, driving in the same direction with, ship2*⟩, ⟨*ship2, driving in the opposite direction with, ship3*⟩, and ⟨*ship1, within safe distance of, ship3*⟩.

Qualitative comparisons on large-extent scenes are presented in Fig. 8. In the upper multidock port example, SARNet produces more accurate predictions for relations within local functional regions, such as ships docking around the same dock area, indicating that the proposed object-centered spatial influence modeling can effectively enhance local relational context. More importantly, SARNet can also recognize relations between objects located in different functional regions. For example, the prediction of ⟨*ship1, docking at the different dock with, ship10*⟩ requires the model to compare the local dock environments around two spatially separated ships, rather than simply relying on their direct distance. This benefits from the combination of global awareness, object-centered local influence modeling, and relation-level spatial configuration refinement, which allows SARNet to suppress irrelevant long-range noise while preserving relation-relevant cross-region dependencies. In the airport example with complex spatial layouts, SARNet also correctly predicts spatially sensitive relations such as ⟨*taxiway10, intersect, runway1*⟩ and ⟨*taxiway12, converge, runway1*⟩, further demonstrating its ability to reason over elongated structures and complex geometric arrangements.

3) Efficiency and Scalability Analysis: For efficiency assessment, we compare different methods on the PredCls task of the STAR test set in Table III. For FLOPs and inference

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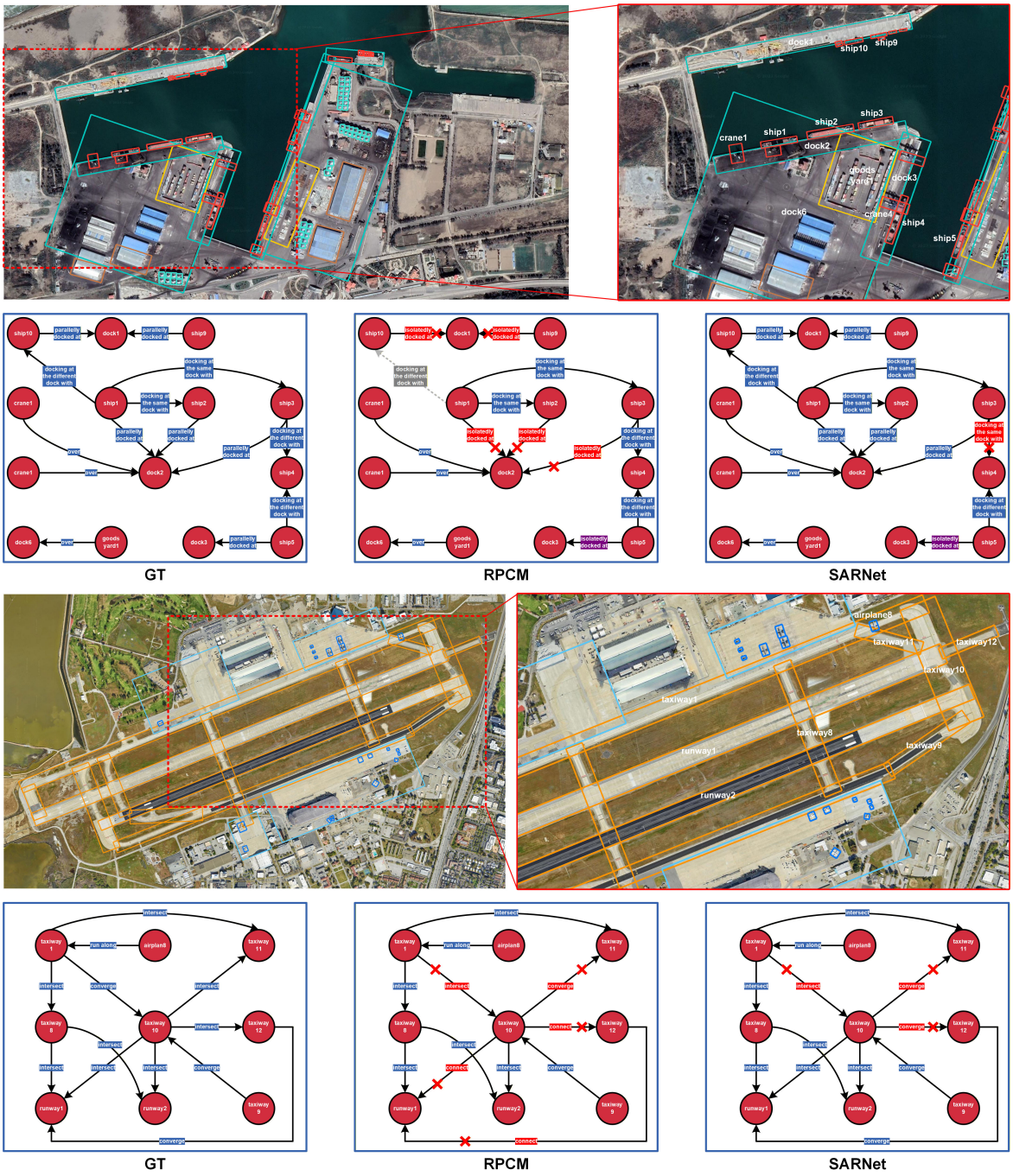


Fig. 8. Visualization results of RPCM and SARNet on large-extent scenes from the STAR test set, including scenarios with multiple subscenes or complex spatial layouts. The edge colors have the same meanings as those in Figs. 6 and 7.

TABLE III

COMPARISON OF MODEL PARAMETERS, FLOPS, AND INFERENCE TIME ON THE PREDCLS TASK ON THE STAR TEST SET

Method	MR/mMR@2000 (%)	Params (M)	FLOPs (G)	Time/img (s)
Motifs[25]	64.00/30.64	48.1	1,682	0.57
HETSGG[32]	65.37/30.98	34.1	1,690	0.83
PENet[59]	64.98/38.29	103.8	2,277	0.61
RPCM[1]	65.86/42.30	236.1	4,249	0.83
SARNet	70.77/40.67	154.2	2,176	0.76

time, we report the cost of the complete PredCls inference process. The reported inference time is averaged over all test images and covers the complete model forward pass with CUDA synchronization, excluding data loading and subsequent evaluation. For model size, we count only the parameters of the SGG components and exclude the shared backbone. Compared with RPCM, SARNet uses fewer SGG parameters (154.2 M versus 236.1 M), requires lower computational cost (2176 G versus 4249 G), and achieves faster inference (0.76 s/image versus 0.83 s/image). These results indicate that SARNet achieves strong relation reasoning performance with a more compact model and lower computational cost.

To further evaluate scalability with respect to object density, we analyze the behavior of SARNet by varying the maximum

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TABLE IV

SCALABILITY ANALYSIS OF SARNET WITH DIFFERENT NUMBERS OF INPUT OBJECTS ON THE PREDCLS TASK OF THE STAR TEST SET

No. of Objects	Memory (MB)	FLOPs (G)	Time/img (s)
5	4,216	1,194	0.12
10	4,216	1,207	0.13
100	6,398	2,053	0.41
2000	13,306	2,176	0.74

TABLE V

SCALABILITY ANALYSIS OF SARNET WITH DIFFERENT NUMBERS OF RETAINED OBJECTS ON THE SGDET TASK OF THE STAR TEST SET

No. of Objects	No. of Relations	Memory (MB)	Time/img (s)
5	14	12,278	9.12
10	59	12,278	9.18
100	3,027	12,278	9.61
2000	4,350	12,278	13.90

TABLE VI

SGDET PERFORMANCE UNDER DIFFERENT SETTINGS OF λ_w^0, λ_h^0, AND γ^0

λ_w^0, λ_h^0	γ^0	SGDet			
		MR@1500	MR@2000	mMR@1500	mMR@2000
1.5	1.0	35.63	36.45	14.57	15.03
1.5	1.5	35.41	36.18	**15.04**	**15.51**
1.5	2.0	35.50	36.29	14.67	15.08
2.0	1.0	35.38	36.20	14.48	14.95
2.0	**1.5**	**35.73**	**36.64**	14.88	15.37
2.0	2.0	**35.73**	36.56	14.69	15.26
2.5	1.0	35.67	36.49	14.59	15.11
2.5	1.5	35.23	36.03	14.62	15.06
2.5	2.0	35.61	36.44	14.48	14.90

number of objects retained from each image in the STAR test set. The PredCls and SGDet results are reported in Tables IV and V, respectively. Memory denotes peak allocated GPU memory measured for each image. Notably, the maximum number of objects per image in the STAR test set is 1842. The last rows of Tables III and IV correspond to the same test conditions, and the runtime difference reflects normal fluctuations. For PredCls, the full detection pipeline is bypassed and only the detector backbone is used for feature extraction, allowing the results to more directly reflect the scalability of the SGG model itself. As the number of retained objects increases, additional candidate pairs and relation features raise the computational cost, which nevertheless remains bounded by the pruning strategy in the standardized inference protocol. For SGDet, object detection on large-area images largely determines the inference time. Because peak GPU memory is dominated by the detector, it remains unchanged across object caps, whereas inference time increases with the number of retained candidate relations.

F. Ablation Study

1) Hyperparameter Analysis of Anisotropic Gaussian Ellipse-based Gate: We analyze the hyperparameters in the anisotropic Gaussian ellipse-based gate of AEIP, including λ_w^0, λ_h^0, and γ^0. λ_w^0 and λ_h^0 control the basic expansion strength of the anisotropic Gaussian ellipse along the width and height axes, respectively, while γ^0 controls the basic decay rate of the influence field. We set $\lambda_w^0 = \lambda_h^0 \in \{1.5, 2, 2.5\}$ and $\gamma^0 \in \{1, 1.5, 2\}$. Results are reported in Table VI.

We observe the best performance when $\lambda_w^0 = \lambda_h^0 = 2$ and $\gamma^0 = 1.5$. This trend supports our intuition that overly broad interactions may introduce additional noise: a larger influence range and stronger interaction are not always beneficial, and local modeling is crucial for reliable predicate prediction in large-area remote sensing scenes.

TABLE VII

SGDET PERFORMANCE AND INFERENCE TIME WITH DIFFERENT NUMBERS OF RSCA LAYERS

Layer	SGDet			
	MR@1500/2000	mMR@1500/2000	HMR@1500/2000	Time/img (s)
L=1	35.43/36.24	14.79/15.35	20.87/21.57	13.66
L=2	35.56/36.33	14.67/15.10	20.77/21.33	13.69
L=3	**36.04/36.86**	14.63/15.16	20.81/21.48	13.79
L=4	35.73/36.64	**14.88/15.37**	**21.01/21.66**	13.90

2) Number of RSCA Layers: We study the number of stacked RSCA layers in the relation decoder. We evaluate the model with 1–4 RSCA layers while keeping all other settings the same. As shown in Table VII, four layers achieve the highest harmonic mean of MR and mMR (HMR [1]), providing the best balance between the two metrics. Meanwhile, inference time increases only slightly as more layers are stacked, from 13.66 s with one layer to 13.90 s with four layers. Considering both performance and computational efficiency, we set the number of RSCA layers to 4.

3) Effectiveness of AEIP and RSCA: We conduct ablation studies in Table VIII to validate the effectiveness of AEIP and RSCA. Table VIII shows that both AEIP and RSCA consistently improve MR@1500, MR@2000, mMR@1500, and mMR@2000 across PredCls, SGCls, and SGDet.

AEIP brings the most substantial gains. For example, on PredCls, mMR@1500/mMR@2000 increases from 33.25%/34.92% to 39.55%/40.58%, showing that explicitly injecting spatial cues into gated message passing strengthens object-centered local context perception and improves relation reasoning reliability. RSCA also yields stable improvements and is more prominent on SGCls, where mMR@1500/mMR@2000 improves from 24.68%/25.53% to 27.24%/28.20%. This suggests that incorporating subject–object–relation geometry into decoding helps characterize complex intrarelation spatial layouts.

Combining AEIP and RSCA produces complementary gains and achieves the overall best performance. The full model is slightly lower than the AEIP-only variant on PredCls mMR@1500 (39.53% versus 39.55%), which may be caused by statistical fluctuation. On SGDet, SARNet achieves 14.88%/15.37% on mMR@1500/mMR@2000, outperforming the single-module variants (13.65%/14.13% and 13.41%/13.89%). These results validate our framework which progresses from global perception to object-centered anisotropic influence propagation and finally to fine-grained spatial-structure reasoning, as an effective formulation for suppressing noisy interactions and improving scene-level relational inference in remote sensing imagery.

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TABLE VIII
ABLATION STUDY OF AEIP AND RSCA ON STAR

Module		PredCls				SGCls				SGDet			
AEIP	RSCA	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000
		65.30	67.29	33.25	34.92	52.14	53.68	24.68	25.53	33.61	34.42	13.24	13.65
✓		68.74	70.45	**39.55**	40.58	**54.77**	**56.02**	26.80	27.50	35.05	35.78	14.04	14.47
	✓	66.98	68.82	36.17	37.38	53.05	54.39	27.24	28.20	33.87	34.75	13.41	13.89
✓	✓	**69.06**	**70.77**	39.53	**40.67**	54.41	55.65	**27.91**	**28.73**	**35.73**	**36.64**	**14.88**	**15.37**

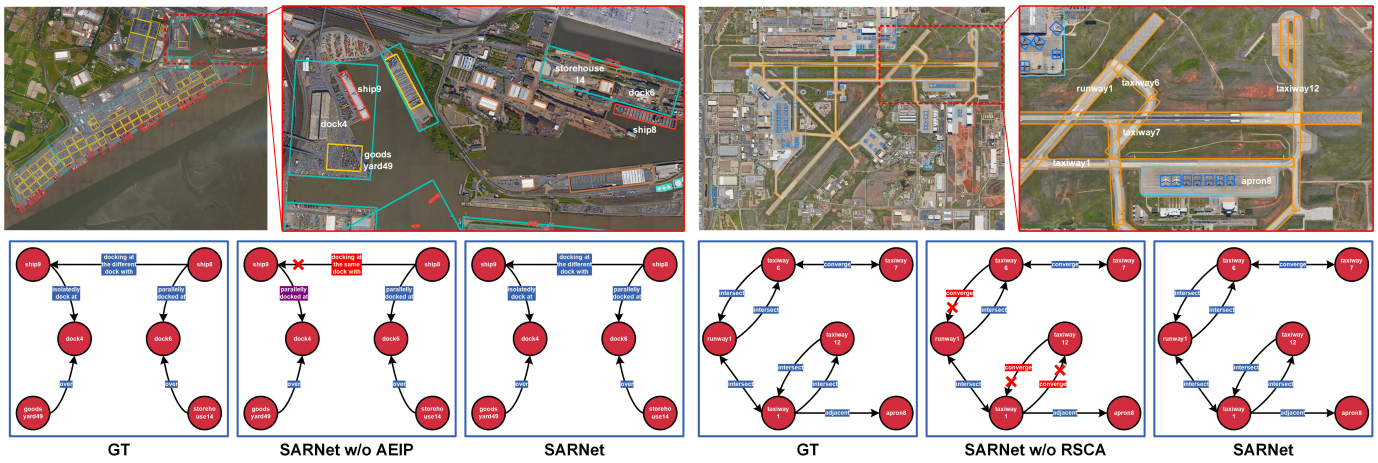


Fig. 9. Qualitative comparison between the ablated variants and the full SARNet on representative STAR test images. The left and right compare the variants after removing AEIP and RSCA, respectively, showing the effects of AEIP and RSCA. The edge colors have the same meanings as those in Figs. 6–8.

TABLE IX
COMPARISON OF PERFORMANCE AND COMPUTATIONAL COST FOR DIFFERENT SARNET VARIANTS ON THE PREDCLS TASK OF THE STAR TEST SET

Method	MR/mMR@2000 (%)	FLOPs (G)	Time/img (s)
–	67.29/34.92	1,695	0.63
AEIP	70.45/40.58	1,865	0.74
RSCA	68.82/37.38	2,007	0.66
SARNet	70.77/40.67	2,176	0.76

We compare different SARNet variants on the PredCls task to evaluate their performance–cost tradeoffs. As reported in Table IX, both AEIP and RSCA improve recognition performance with moderate increases in FLOPs and inference time, demonstrating that the proposed modules achieve favorable efficiency–effectiveness tradeoffs.

We further visualize the effects of AEIP and RSCA through separate ablated variants in Fig. 9. In the left example, correctly predicting ⟨*ship8, docking at the different dock with, ship9*⟩ requires the model to perceive and compare the local docking environments around ship8 and ship9, where dock4 and dock6 indicate that the two ships are berthed at different dock regions. After removing AEIP, the object representations insufficiently encode such relation-relevant local context, causing the model to incorrectly predict ⟨*ship8, docking at the same dock with, ship9*⟩. Moreover, although ⟨*ship8, parallelly docked at, dock4*⟩ is semantically reasonable, the full SARNet produces the more precise prediction ⟨*ship8, isolatedly dock at, dock4*⟩ by confirming that no other ship is docked around dock4. In the right example, the intertwined topology among runways, taxiways, and aprons makes the predicates "converge" and "intersect" easily confused. By capturing the relative spatial configuration between the subject and object, SARNet avoids the geometry-sensitive misclassification observed after removing RSCA, and correctly predicts relations such as ⟨*taxiway6, intersect, runway1*⟩.

To validate the effectiveness of object-centered anisotropic local interaction modeling, we conduct an ablation study on the anisotropic Gaussian ellipse-based gating process in AEIP (denoted by *GaussGate*). To avoid confounding factors and more clearly isolate the contribution of GaussGate, we remove RSCA from SARNet and compare only between the baseline and AEIP variants. As reported in Table X, introducing GaussGate into the relation-to-object context flow consistently improves performance across all metrics for the three tasks, with particularly notable gains in recall for PredCls and SGCls.

Notably, GaussGate only introduces a small number of learnable parameters that generate instance-wise adaptive offsets. Moreover, compared with a full MPNN, AEIP with GaussGate achieves higher accuracy despite attenuating the magnitude of relation messages along edges with weak spatial relevance. This result highlights the importance of leveraging object spatial cues to construct anisotropic, object-centered influence regions for selective context aggregation.

Similarly, we further examine the impact of explicitly modeling the relative spatial configuration of the subject and object within each relation. Following a design analogous to the previous ablation, we remove AEIP to avoid additional confounding factors and focus on isolating the role of subject–object and relation-region geometry by comparing the

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TABLE X
ABLATION STUDY ON ANISOTROPIC GAUSSIAN ELLIPSE-BASED GATING

Component		PredCls				SGCls				SGDet			
Message Passing	GaussGate	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000
		65.30	67.29	33.25	34.92	52.14	53.68	24.68	25.53	33.61	34.42	13.24	13.65
✓		67.70	69.53	39.13	40.32	52.91	54.21	26.51	27.19	34.99	**35.78**	13.78	14.17
✓	✓	**68.74**	**70.45**	**39.55**	**40.58**	**54.77**	**56.02**	**26.80**	**27.50**	**35.05**	**35.78**	**14.04**	**14.47**

TABLE XI
EFFECTIVENESS OF RELATIVE GEOMETRIC CONFIGURATION, WHICH IS DENOTED BY RELGEO. SIMATTN DENOTES THE SIMPLIFIED ATTENTION LAYER

Component		PredCls				SGCls				SGDet			
SimAttn	RelGeo	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000
		65.30	67.29	33.25	34.92	52.14	53.68	24.68	25.53	33.61	34.42	13.24	13.65
✓		65.97	67.85	36.12	**37.44**	51.96	53.47	25.70	26.56	33.11	33.94	12.80	13.18
✓	✓	**66.98**	**68.82**	**36.17**	37.38	**53.05**	**54.39**	**27.24**	**28.20**	**33.87**	**34.75**	**13.41**	**13.89**

TABLE XII
PERFORMANCE AND EFFICIENCY COMPARISON OF DIRECT GEOMETRIC FUSION AND SPATIAL-CONFIGURATION-GUIDED FEATURE REALLOCATION ON THE PREDCLS TASK OF THE STAR TEST SET

Method	MR@1500/2000 (%)	mMR@1500/2000 (%)	FLOPs (G)	Time/img (s)
–	65.30/67.29	33.25/34.92	1,695	0.63
CatGeo	66.37/68.24	34.55/35.83	1,789	0.65
RSCA	66.98/68.82	36.17/37.38	2,007	0.66

baseline with RSCA variants. As a comparison, we remove all implementations related to the relative geometry configuration injection, which degrades to a simplified attention layer (denoted by SimAttn). As shown in Table XI, simply refining relation features by attending to the subject and object features, without injecting spatial cues, provides limited gains. This is largely due to the complexity of remote sensing scenes, where heavy clutter and noise can easily obscure semantic matching. By contrast, incorporating explicit subject–object–relation geometry to model the within-relation relative layout leads to clear improvements, resulting in better performance across all three tasks.

Besides, to further assess the contribution of the attention-based design in RSCA, we introduce a simpler geometric fusion baseline, termed CatGeo, which directly concatenates the geometric encodings of the subject and object with the relation feature. We compare CatGeo with RSCA in terms of recognition performance, computational cost, and inference efficiency in Table XII. CatGeo improves the baseline with moderate additional FLOPs and inference time, confirming the usefulness of subject–object geometry for relation prediction. However, its gains are consistently lower than those of RSCA. Although RSCA incurs more FLOPs, its additional runtime overhead over CatGeo is marginal, while it achieves stronger MR and mMR performance. This indicates that RSCA more effectively exploits relative spatial configurations through relation-specific feature reallocation, rather than simply incorporating geometric cues.

G. Exploration on the Learned Instance-Wise Adaptive Offsets

We analyze the distribution of the learned adaptive offsets ($\Delta\lambda_{w,i}$, $\Delta\lambda_{h,i}$, and $\Delta\gamma_i$) of the anisotropic Gaussian ellipses for several representative object categories that frequently appear in remote sensing scenes. We select three typical types of objects: small rigid objects with stable shapes (e.g., car and boat), region-level functional areas with relatively fixed shapes (e.g., car parking and goods yard), and large-area regions with significant shape variations (e.g., dock and taxiway).

From Fig. 10, the following observations and conclusions can be drawn.

For the effective expansion offsets $\Delta\lambda_{w,i}$ and $\Delta\lambda_{h,i}$, the distributions of car, boat, dock, and taxiway exhibit wide ranges, while car parking and goods yard show highly concentrated and stable values. The large variation of dock and taxiway mainly comes from their diverse shapes and scales across different scenes, which requires a wider adaptive adjustment of the influence region. Although car and boat have relatively stable shapes, their spatial distributions vary greatly in remote sensing images, including dense parking, sparse placement, and moving states, leading to different interaction ranges with surrounding objects and thus larger offset variations. In contrast, car parking and goods yard usually appear as regular functional areas with consistent spatial layouts, resulting in more stable offset distributions.

For the decay offset $\Delta\gamma_i$, car parking, goods yard, dock, and taxiway generally have smaller values than car and boat, indicating slower attenuation over a larger Gaussian ellipse, which forms region-level influence. In contrast, small objects tend to have smaller ellipses with faster decay, leading to more localized influence. This behavior is consistent with intuitive expectations that large functional regions affect wider surroundings, while small objects mainly interact within local neighborhoods.

As shown in Fig. 11, we visualize the anisotropic Gaussian ellipse influence regions of car, goods yard, and taxiway from left to right. The influence strength is presented as heatmaps, where the intensity decays with increasing distance from the object center.

For car, the influence region is concentrated in the local neighborhood, with diffusion along the major axis effectively constrained, resulting in a compact spatial interaction range. For goods yard, the diffusion is relatively balanced along both axes, producing a wider and more uniform influence region

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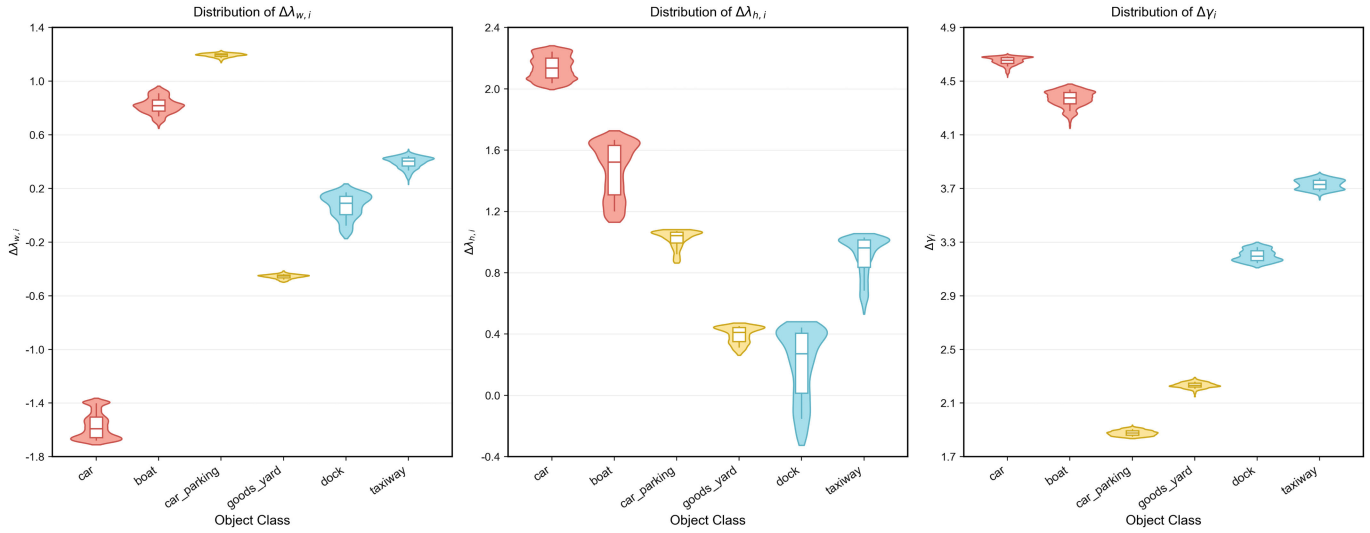


Fig. 10. Distribution of the learned adaptive offsets $\Delta\lambda_{w,i}$, $\Delta\lambda_{h,i}$, and $\Delta\gamma_i$ for three typical types of objects.



Fig. 11. Visualization of the learned anisotropic Gaussian ellipse for the car, goods yard, and taxiway.

that covers the surrounding environment and related facilities. In contrast, taxiway exhibits a highly elongated influence region, where the diffusion along the major axis extends over a large spatial extent, leading to long-range influence consistent with its elongated geometry.

H. Exploration of the GaussGate

To verify the independent contribution of GaussGate and disentangle the effects of its internal designs, we conduct an additional gate-strategy ablation within the complete SARNet framework. The compared variants are organized from five perspectives.

1) We introduce a no-gate variant, which removes explicit gate modulation while keeping the message-passing pipeline unchanged, to evaluate the effect of gate-controlled relation-to-object propagation.
2) We consider simple spatial-neighborhood gates, denoted as TOP-2 and TOP-4, where each object only receives messages from its top-K nearest neighboring objects according to center distance. These variants evaluate whether simple locality constraints are sufficient for RS-SGG.
3) We compare with RCE [31], a confidence-based nonuniform weighting strategy originally designed to suppress false-positive relation messages during bipartite message passing. This comparison helps distinguish the proposed geometry-aware spatial gate from the general benefit of introducing nonuniform message weights.
4) We introduce PairMLP, a generic learnable pairwise scalar gate evaluated under the same SARNet settings and using the same geometric and soft semantic inputs as GaussGate. It constructs a symmetric pair representation by concatenating the feature sum and absolute difference, and directly maps this representation to a scalar gate through an MLP. This design enables a controlled performance comparison with GaussGate.
5) We design four controlled variants of GaussGate to analyze its internal components. The w/o Adapt. variant disables all instance-wise adaptive offsets. The w/Adapt. Exp. and w/Adapt. Decay variants retain only the adaptive expansion offsets and the adaptive decay offset, respectively. These variants separate the effects of adaptive influence-range adjustment and adaptive distance decay. The circular variant replaces the anisotropic elliptical influence region with an isotropic circular one to evaluate anisotropic spatial modeling.

The results reported in Table XIII show that GaussGate consistently improves the no-gate variant across the three SGG

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TABLE XIII
ABLATION STUDY OF DIFFERENT GATE STRATEGIES ON STAR

Gate	PredCls				SGCls				SGDet			
	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000
–	68.07	69.79	39.02	40.05	53.85	55.09	27.52	28.13	34.85	35.75	14.47	14.96
TOP-2	68.59	70.19	38.54	39.65	54.29	55.49	27.62	28.45	34.90	35.95	14.40	15.03
TOP-4	68.77	70.35	39.10	40.06	**54.56**	55.77	27.47	28.09	34.59	35.65	14.64	15.22
RCE[31]	68.21	69.93	39.37	40.45	54.07	55.37	26.92	27.61	35.28	36.17	14.42	14.90
PairMLP	67.77	69.63	39.40	40.56	53.45	54.65	27.58	28.22	35.15	36.03	14.66	15.15
w/o Adapt.	68.78	70.43	38.69	39.99	54.51	**55.78**	27.27	28.01	35.22	35.95	14.85	15.29
Circular	67.82	69.38	38.93	39.89	53.62	54.85	26.98	27.57	35.07	35.96	14.57	14.96
w/ Adapt. Exp.	68.88	70.61	38.95	40.01	54.19	55.44	27.52	28.28	35.62	36.47	14.46	15.04
w/ Adapt. Decay	**69.35**	**71.12**	38.47	39.77	54.48	55.73	27.66	28.31	35.36	36.11	14.49	14.90
GaussGate	69.06	70.77	**39.53**	**40.67**	54.41	55.65	**27.91**	**28.73**	**35.73**	**36.64**	**14.88**	**15.37**

TABLE XIV
MEAN ± STANDARD DEVIATION OF KEY GATE VARIANTS OVER THREE REPEATED RUNS ON STAR

Gate	PredCls				SGCls				SGDet			
	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000	MR@1500	MR@2000	mMR@1500	mMR@2000
TOP-4	69.01 ± 0.20	70.60 ± 0.22	39.38 ± 0.20	40.45 ± 0.28	54.45 ± 0.14	55.67 ± 0.17	27.10 ± 0.41	27.78 ± 0.40	34.70 ± 0.12	35.71 ± 0.08	14.88 ± 0.18	15.39 ± 0.12
GaussGate	68.94 ± 0.13	70.73 ± 0.10	39.86 ± 0.24	41.04 ± 0.27	54.81 ± 0.47	56.05 ± 0.42	27.98 ± 0.05	28.79 ± 0.04	35.79 ± 0.24	36.63 ± 0.25	15.00 ± 0.08	15.46 ± 0.07

tasks, demonstrating the effectiveness of explicitly regulating relation-to-object propagation. Among all gate strategies, GaussGate achieves the best overall performance, obtaining the highest scores on all SGDet metrics and the strongest mMR performance on both PredCls and SGCls. Given the relatively small performance differences among the gate strategies, we select TOP-4 as a representative strategy based on its competitive performance and conduct three training runs for both TOP-4 and GaussGate to assess performance stability. As reported in Table XIV, SARNet with GaussGate exhibits limited variation across runs and achieves higher mean scores than TOP-4 on most metrics. These results further support the overall advantage of GaussGate under the evaluated settings.

TOP-2 and TOP-4 improve over the no-gate setting in some cases, but their gains are unstable across different tasks. This indicates that simply restricting propagation to spatially nearest neighbors is insufficient. In remote sensing scenes, center-to-center distance can be unreliable for objects with elongated shapes or large aspect ratios: two objects may be physically adjacent, parallel, or intersecting while their centers remain far apart. Therefore, local interaction modeling requires more than nearest-neighbor selection based on object centers.

The comparison with RCE further shows that generic nonuniform weighting alone cannot fully explain the gain of GaussGate. Although RCE suppresses unreliable relation messages through confidence-based weights, it does not explicitly characterize the geometry-aware spatial influence between objects. In contrast, GaussGate derives relation-to-object propagation weights from object shape, orientation, and spatial extent, providing an interpretable spatial prior for relation-context aggregation. It should be noted that the anisotropic Gaussian ellipse is not intended to exactly reconstruct complex physical layouts such as road topology, shorelines, waterways, or parking boundaries. Instead, it serves as a lightweight, annotation-free, object-centered approximation of spatial influence based on the available OBB geometry. This design provides a practical balance between geometric expressiveness, generality, and efficiency under current RS-SGG annotations. Nevertheless, this also means that GaussGate may not fully capture highly nonelliptical or topology-constrained interactions. More explicit topology- or layout-aware priors could be beneficial when additional structural annotations or reliable layout extraction methods are available, which we leave for future work.

The decoupled variants show that introducing adaptive offsets generally improves overall performance compared with w/o Adapt. However, w/Adapt. Exp. and w/Adapt. Decay individually provide limited gains, whereas combining both components in GaussGate achieves stronger overall performance, showing that jointly adapting the expansion and decay of each object's influence region improves gate construction. In remote sensing scenes, the effective influence range of an object is not solely determined by its OBB size, but also depends on its semantic category, surrounding object density, scale variation, and local spatial context. For example, two ships with similar OBB sizes may still require different influence extents under different local layouts, such as crowded docking areas and open wharf regions, where the density of neighboring objects and the spatial arrangement of dock facilities can substantially change the relation-relevant context. By adaptively adjusting the expansion factors for each instance, GaussGate can flexibly estimate object-specific influence regions and construct more reliable propagation weights.

Under the same nonadaptive setting, w/o Adapt. and Circular mainly differ in the shape of the influence region: the

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TABLE XV
SGDET PERFORMANCE OF SARNET WITH CONFIDENCE-CONSTRAINED PROPAGATION IN AEIP

Gate	MR@1500	MR@2000	mMR@1500	mMR@2000
Conf-weighted	35.53	36.41	14.41	14.84
Conf-thresholded	35.18	36.05	14.46	15.02
GaussGate	**35.73**	**36.64**	**14.88**	**15.37**

former retains an anisotropic ellipse, whereas the latter uses an isotropic circle. The clear advantage of w/o Adapt. over Circular demonstrates that anisotropic modeling is important for RS-SGG, since oriented objects in remote sensing images often have strong directionality and highly variable aspect ratios. By aligning the influence region with the major and minor axes of OBBs, GaussGate captures object spatial influence in a more fine-grained and orientation-aware manner.

Overall, these comparisons show that the improvement of GaussGate comes not merely from adding any weighted propagation mechanism, but from geometry-aware, anisotropic, and adaptively adjusted spatial influence modeling.

I. Impact of False-Positive Proposals on AEIP

Since AEIP enhances object representations through gated relation-context propagation, nearby false-positive proposals may also receive nonnegligible GaussGate weights under the highly noisy SGDet setting. We therefore investigate whether explicitly constraining GaussGate by detection confidence can further improve robustness against false-positive detections. Specifically, we design two GaussGate-based confidence-constrained variants: *Conf-weighted* multiplies each GaussGate weight by the detection confidence of the message-providing object, while *Conf-thresholded* permits GaussGate propagation only from proposals with confidence higher than 0.6.

As shown in Table XV, both confidence-constrained variants degrade SGDet performance compared with the original GaussGate. This indicates that directly suppressing low-confidence proposals during AEIP propagation is not an effective strategy. The reason lies in the bipartite object–relation propagation design of AEIP. GaussGate mainly controls the relation-to-object context flow, where the propagated message is a relation-level representation extracted from the union region of an object pair, rather than an isolated object feature. Such relation features can encode useful local contextual cues, including surrounding facilities, functional layouts, and object density. Therefore, even when a proposal is counted as a false positive during evaluation, its associated relation feature may still provide contextual evidence for neighboring objects. Consequently, AEIP does not aim to remove false-positive detections at the object level. Instead, it improves SGDet robustness by regulating relation-level context aggregation within geometry-consistent local influence regions. The superior performance of the original GaussGate over the confidence-constrained variants suggests that overly conservative confidence-based filtering may discard informative contextual cues and weaken object-centered local context modeling.

V. CONCLUSION

In this work, we design SARNet for RS-SGG under large-area remote sensing scenes with coexisting subscenes and highly complex object spatial layouts. Specifically, AEIP adaptively models the spatial influence region of each object and embeds spatial priors into object interactions, thereby propagating locally coherent context. Meanwhile, RSCA leverages relative geometric configurations to guide attention allocation for emphasizing discriminative spatial cues, thereby incorporating subject–object spatial structure into relation representations. Our method effectively improves the prediction of predicates that are highly correlated with surrounding context and spatial distributions, and the experimental results verify its superior performance. We believe that, in RS-SGG, scene decoupling is a key factor for improving predicate prediction and also a promising direction for further exploration and optimization. Meanwhile, we observe that existing models generally perform poorly in complex dynamic traffic scenes. This may be attributed to the lack of visual information for small objects in traffic scenes, as well as the difficulty of capturing the semantic regularities of some related predicates, suggesting that additional dedicated designs may still be needed.

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